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**Update of the Stochastic Production model in Continuous Time (SPiCT) applied
to *Dosidicus gigas* in the FAO area 87**

Chile



Update of the Stochastic Production model in Continuous Time (SPiCT) applied to *Dosidicus gigas* in the FAO area 87.

By Ignacio S. Payá C.

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Stock Assessment Department. Chilean Fisheries Development Institute.

ABSTRACT

Because in-season stock assessment and management are difficult to implement for *Dosidicus gigas* in the SPRFMO area, an annual stock assessment using the Stochastic Production Model in Continuous Time (SPiCT) is a viable alternative. The aim of this study was to update the squid stock assessment using SPiCT with data up to 2024, representing a significant improvement over previous assessments that relied on data with a two-year delay.

A total of six CPUE abundance indices from different countries were used in the SPiCT model. The updated CPUE index for the Chinese fleet was provided by Chinese scientists. A sensitivity analysis was conducted to evaluate the impact of fitting the model to a single global abundance index versus country-specific indices, fixing the Schaefer model, and applying a prior distribution for the parameter r .

A significant reduction (52%) in total squid catch occurred in 2024, and a general decreasing trend was observed in the abundance indices. Biomass showed an increasing trend from 2001 to 2010, followed by a decline. The overall stock trajectory in the Kobe plot was similar across all cases, but the estimated 2025 stock status varied: four cases fell into the red zone, one into the yellow, and two into the green zone, all with considerable uncertainty.

Estimates of population growth (r) and carrying capacity (K) varied widely across cases and were negatively correlated, resulting in an MSY of approximately 950 kt. The estimated TAC for 2026 under FMSY ranged from 267 kt (Case 6: Schaefer model with six indices and r prior) to 989 kt (Case 1: productivity curve with r freely estimated and one global index). The case with the best retrospective pattern (Case 7: productivity curve freely estimated with six indices and r prior) estimated a 2025 stock status in the red zone and a TAC for 2026 of 467 kt.

Additionally, the impact of fitting SPiCT to a shorter time series (2012–2024) was analyzed. Two of the seven cases did not converge; four estimated a 2025 status in the red zone, and one in the yellow zone. The MSY (941–950 kt) and TAC 2026 (271–756 kt) were similar to those estimated using the full time series (2001–2024). Further analyses are needed on abundance indices, phenotypic effects, and environmental variability



BACKGROUND

Recommendations for assessing short-lived squid stocks include depletion models with multiple recruitment pulses (Payá 2009; Arkhipkin et al. 2020). However, fitting depletion models requires a fine temporal scale to correctly identify depletion events and the arrival of new recruitment pulses. For example, depletion models applied to *Doryteuthis gahi* in the Falkland Islands have shown that daily time resolution is necessary to detect new recruitment arrivals; otherwise, these may be confused with previous depletion events within a week's time (Payá 2009).

Depletion models with multiple pulses have also been applied to *Dosidicus gigas* fisheries targeting large phenotypes in Chile, using weekly data over several years (Payá 2018a). It has been proposed to the SPRFMO Scientific Committee to apply such models to the SPRFMO area and coastal country fisheries, integrate the results by country into a production model, and test this approach using a squid simulation model (Payá 2018b, 2018c).

So far, the characteristics of *D. gigas* fishing operations in the SPRFMO area, along with current monitoring and biological sampling practices, make it difficult to obtain data at daily or weekly resolution. Furthermore, in-season stock assessment and management appear to be challenging to implement for *D. gigas* in this region.

An alternative method for squid stock assessment could be a surplus production model in continuous time, known as SPiCT (Pedersen and Berg 2017), which allows assessments by month or quarter. SPiCT is currently used to assess several stocks within ICES (ICES 2021).

The first attempt to apply SPiCT to *D. gigas* in FAO Area 87 (Figure 1) assumed a single stock, used annual data, and included a prior distribution for the intrinsic population growth rate (Payá 2022). The model fit the data well using a global abundance index. When fitted to country-specific indices, the model converged, but the retrospective analysis failed. Although different production models estimated similar biomass levels, the growth rate (r) varied. BMSY was estimated at 2,521 thousand tons, and MSY at 728 thousand tons. Stock status was highly uncertain, but the stock appeared to be overfished.

The second and third SPiCT applications (Payá 2023, 2024) aimed to evaluate the impact of fitting the model to a global versus country-specific abundance index, fixing the Schaefer model, using more informative priors for r , and changing the initial year of the data series. More informative r priors were defined using values from the same or similar squid species. Seven cases were analyzed to assess the effects of different abundance indices, productivity curves, and r priors. Estimation of the population growth parameter improved with the use of r priors and adjustments to the initial year.

The most recent assessment found no significant retrospective patterns in biomass or fishing mortality (F), although $B/BMSY$ and $F/FMSY$ did show patterns. Stock status remained highly uncertain and appeared to be overfished. The estimated TAC for 2023 at FMSY ranged from 473,000 tons (Case 5) to 636,000 tons (Case 6).

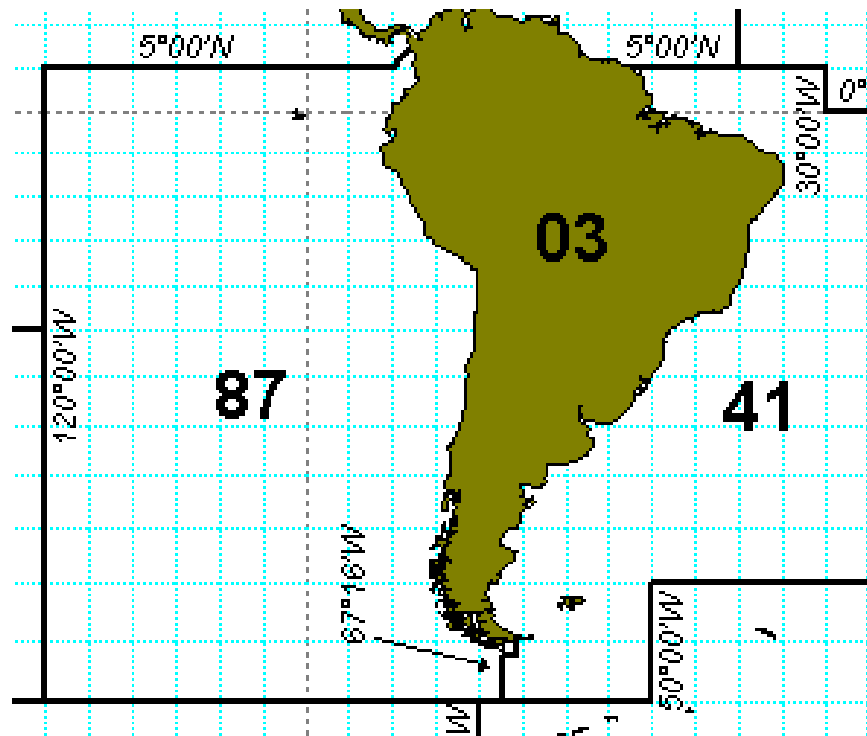


Figure 1. FAO areas. Area 87 includes SPFRMO area; Ecuador EEZ; Perú ANJ; and Chile EEZ.

OBJECTIVES

1. To update the application of the Stochastic Production model in Continuous Time (SPiCT) to *D. gigas* in the FAO87 area.
2. To evaluate the impact of fit the model to one global abundance index or to several indices by country.
3. To evaluate the impact of fixed Schafer model and use a prior distribution for the intrinsic growth rate of the population (r).

DATA AND METHODS

Catches

The catch data by country were collected from SPRFMO statistics and PRODUCE of Perú. A significant reduction (52%) in total squid catch occurred in 2024 (Figure 2).

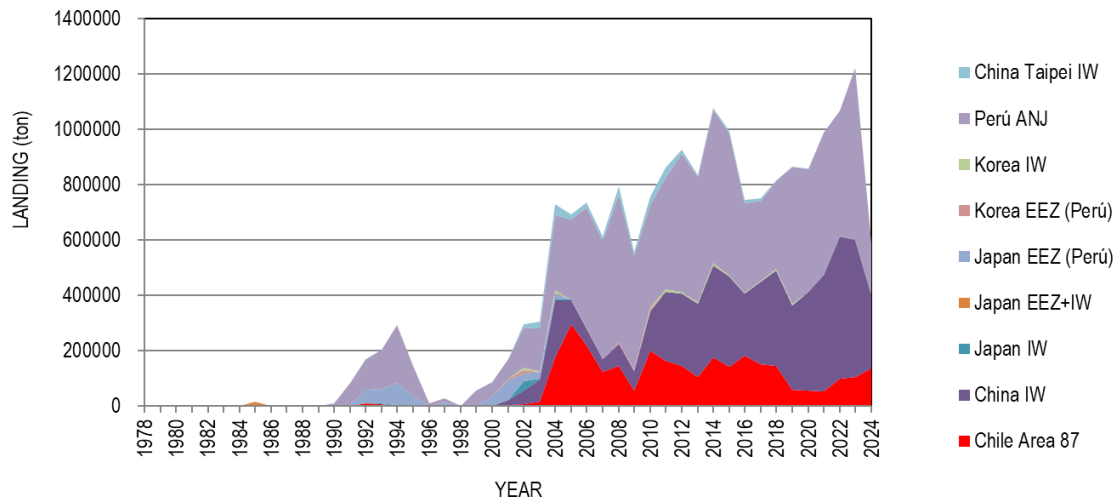


Figure 2. Catch by country.

Abundance indices in SPRFMO area

Two CPUE abundance indices based on Chinese fisheries were available: 1) an index for the 2003–2015 period, based on the year effect from the CPUE model of Xu et al. (2017), and 2) an index for the 2012–2024 period, provided by Dr. Gang Li, derived from the updated CPUE model used in 2024 (Li et al., 2024).

Two scenarios were considered: 1) using the two indices separately, and 2) Combining the two indices into a single CPUE index by scaling Index 1 using the mean ratio (1.76) observed during the overlapping period (2012–2015) (Figure 3).

No CPUE models are currently available for the Chinese Taipei and Korean fleets. Therefore, nominal CPUE values (tons per fishing day) were used as abundance indices, compiled from data available in their national reports (Figure 4).

Abundance index in Chilean EEZ

An abundance index based on CPUE of artisanal boats was used. The CPUE was calculated as ton per fishing trip. The abundance index was the year effect of Linear Mixed Model (LMM) with year, month and region as fixed effects, and boat and interaction Region and Month as random effects of the intercept. A cubic transformation was used to normalized the residuals.

$$CPUE^{1/3} \sim Year + Month + Region + (1 | Boat + Region:Month)$$

Abundance in Peruvian waters

Because there is not abundance index based on CPUE model for the Peruvian fleet, the artisanal CPUE weighted by fishing area following Gulland (1971) method was used (IMARPE 2025). Because no tables were included in the IMARPE report, the CPUE (t/day) were approximated by digitalization of report plots.



Figure 3. Chinese fleet CPUE indices (top), scaled CPUE indices (middle), and combined CPUE index (bottom).

Global abundance index

A global annual index was estimated as an average of relative indices by country weighted by their annual catches. Relative indices were calculated dividing the indices by their values in year 2015.

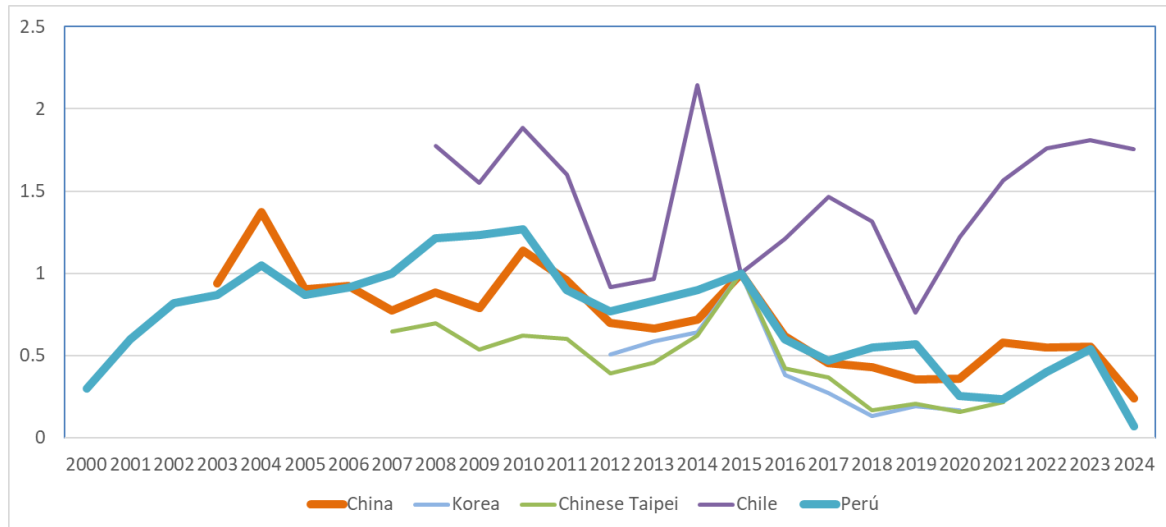


Figure 4. Relative abundance indices by country (Chinese is a combined CPUE index).

Surplus Production model in Continuous Time (SPiCT)

SPiCT is a stochastic production model in continuous time (Pedersen & Berg, 2017) based on generalised surplus production models in the form of Pella & Tomlinson (1969). It is a state-space model that separate random variability of stock dynamics from error in observed indices of biomass. It enables error in the catch process to be reflected in the uncertainty of estimated model parameters and management quantities. It has the ability to provide estimates of exploitable biomass and fishing mortality at any point in time from data sampled at arbitrary and possibly irregular intervals. The biomass and fishing mortality are unobserved processes (random effects), and the catches and abundance indices are observed variables.

Checklist of SPiCT assessment

A SPiCT stock assessment is accepted if it passes the following check list (Mildenberger et al. 2021):

1. The assessment convergence.
2. Variance parameters of the model parameters are finite.
3. No violation of model assumptions based on one-step-ahead residuals (bias, auto-correlation, normality).



4. Consistent patterns in the retrospective analysis.
5. Realistic production curve (The shape of the production curve should not be too skewed (BMSY/K) should be between 0.1 and 0.9).
6. The main variance parameters (logsdb, logsdc, logsdi, logsdf) should not be unrealistically high.

SPiCT is implemented in a R package called “SPiCT” which is based on Template Model Building (TMB, Kristensen et al. 2016).

ICES harvesting rule and BAC in SPiCT

ICES (2021) apply SPiCT for short-life stock assessments. The biological reference points (BRP) are the fishing mortality (F_{MSY}) and biomass (B_{MSY}) at Maximum Sustainable Yield (MSY). ICES recommends a hockey-stick harvesting rule like: $B_{trigger} = B_{MSY} / 2$ and $Blim = B_{MSY} / 3$. The biological acceptable catch (BAC) recommended is the 35th percentile of distribution of catch of short projections.

Sensitivity analysis of SPiCT fits to *D. gigas* in the FAO area 87.

The normal prior distributions for the population growth parameter, r , were the same as defined by Payá (2024), which were based on data in table 1, and described in table 2. Seven cases were defined for analysing the effect of different abundance indices, productive curve and growth parameters (Table 2).

The input data used in cases 1, 3, y 5 are shown in the figure 5, while the input data for cases 2, 4, 6 y 7 in the figure 5. The year 2000 was excluded from the data as the initial year, the fishing effort was very low during that year, and its inclusion in the data had a strong effect on r parameter estimations (Payá 2023).

Sensitivity analysis to shorter time series

To evaluate the effects of shorter the time series, the seven cases were fit with 2012-2024 time series.



Table 1. Growth parameters for the same or similar squid species

r	Species	Area	Reference
1.19	<i>Ommastrephes bartramii</i>	North Pacific	Ichii et al., 2006
0.85	<i>Illex illecebrosus</i>	North Atlantic	Ichii et al., 2006
0.53 (0.16-1.71)	<i>Illex coindetii</i> and <i>Todaropsis eblanae</i>	Northwest Iberian Peninsula	Larivain et al., 2021
1.71-1.90	<i>Ommastrephes bartramii</i>	Northwest Pacific	Wang et al. 2016
1.23 – 1.68	<i>D. gigas</i>	México	Urías-Sotomayor et al., 2018
1.33	<i>D. gigas</i>	Perú	Csirke et al., 2015

Table 2. Description of cases. Prior distribution as normal distribution (mean, std).

	Case 1	Case 2	Case 3	Case 4	Case 5	Case 6	Case 7
Abundance Indices	Global Index (average of country weighted by catch)	6 Indices. 1. China 1 2. China 2 3. Korea 4. China Taipei 5. Chile 6. Perú	Global Index (average of country weighted by catch)	6 Indices. 1. China 1 2. China 2 3. Korea 4. China Taipei 5. Chile 6. Perú	Global Index (average of country weighted by catch)	6 Indices. 1. China 1 2. China 2 3. Korea 4. China Taipei 5. Chile 6. Perú	6 Indices. 1. China 1 2. China 2 3. Korea 4. China Taipei 5. Chile 6. Perú
Shape of productive curve ("parameter n")	Free	Free	Free	Free	Schaefer Model	Schaefer Model	Free
Parameter "r"	Free	Free	Prior (1, 0.6)	Prior (1, 0.5)	Prior (1, 0.5)	Prior (1.5, 0.15)	Prior (1.5, 0.15)

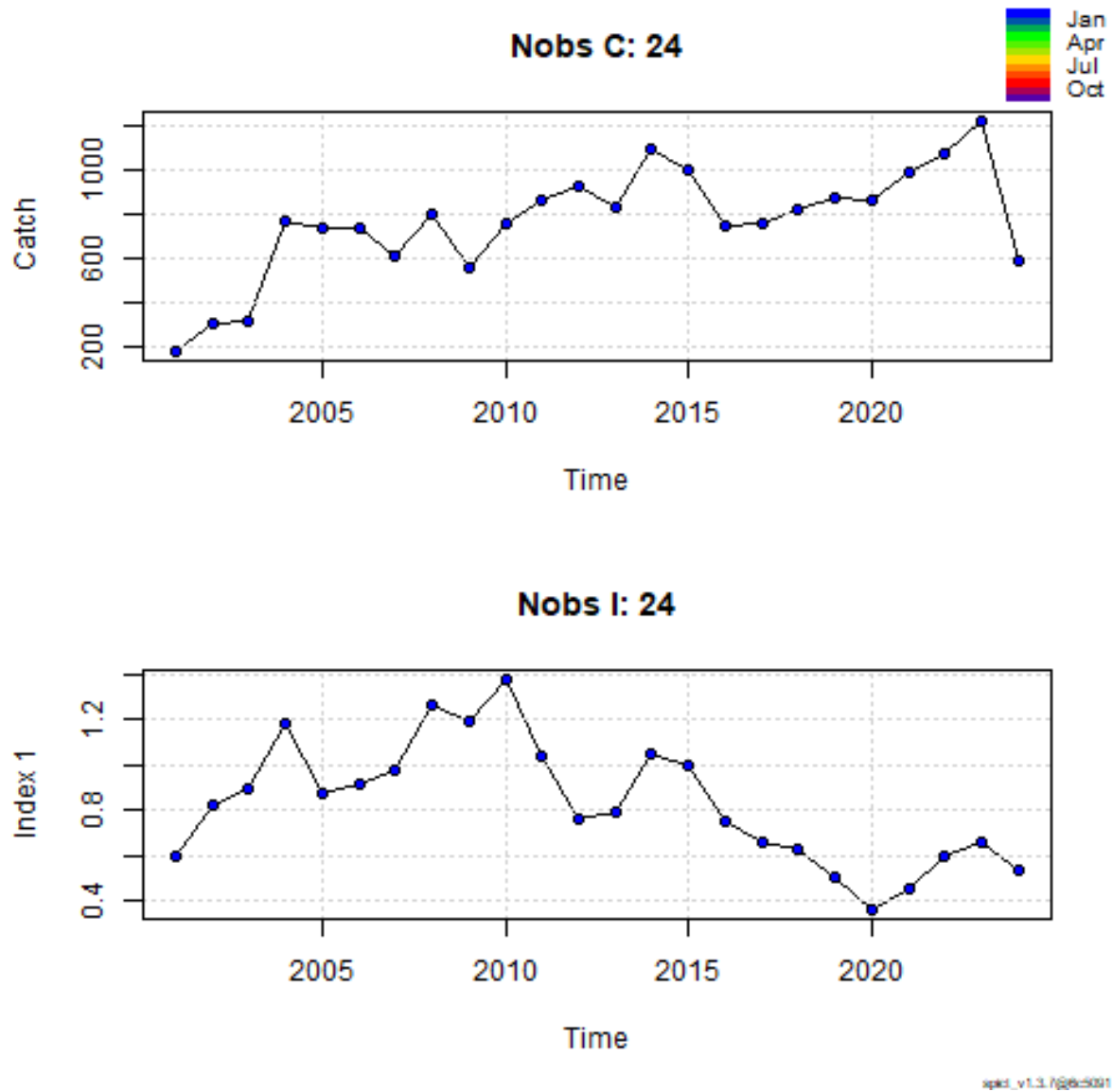
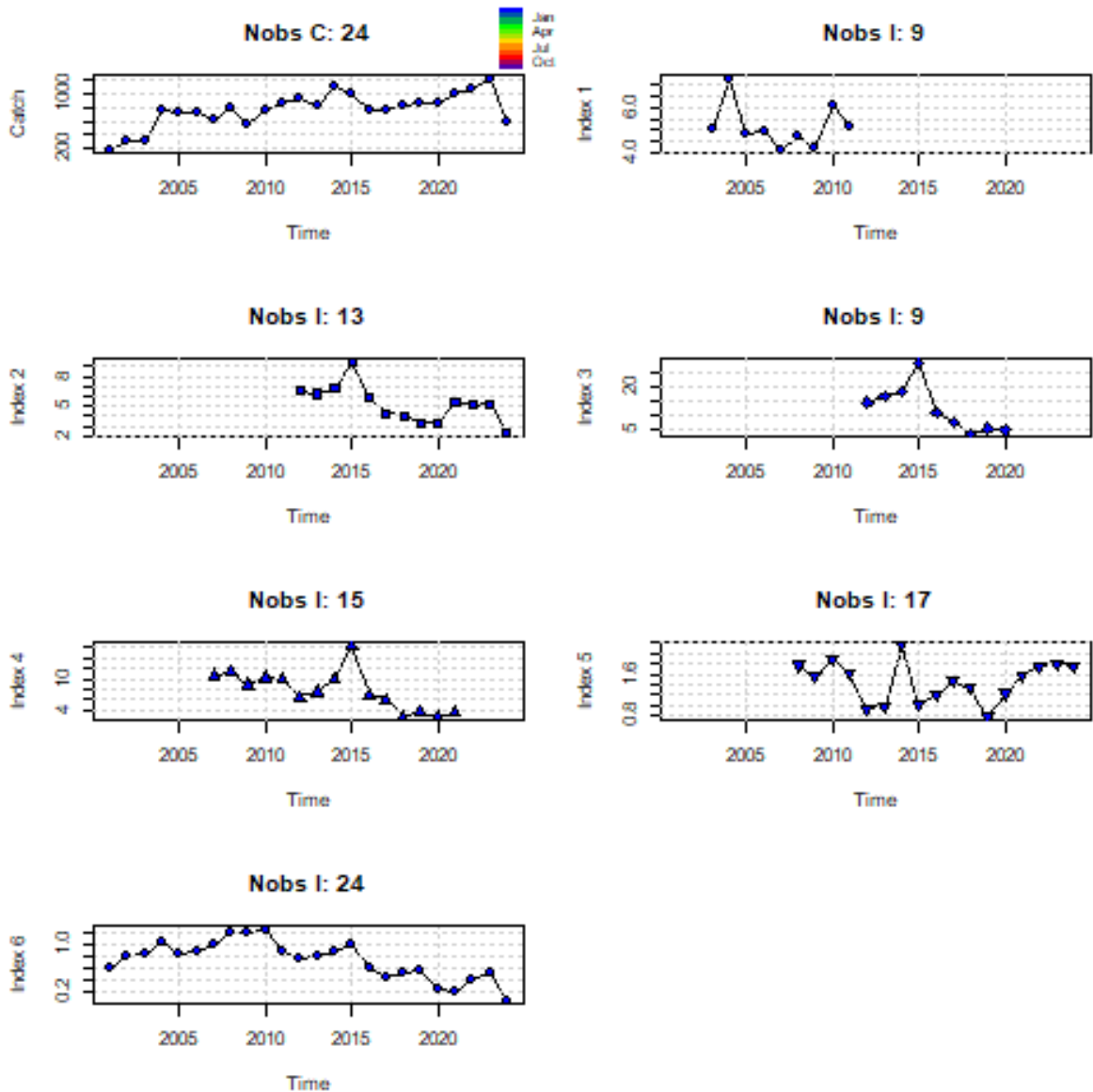


Figure 4. Input data used in cases 1, 3, and 5 (C: Catch; I: Global abundance index).



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Figure 5. Input data used in cases 1, 3, and 5. “I” from left to right and top to down correspond to the abundance indices of China 1, China 2, Korea, China Taipei, Chile and Peru.

RESULTS.

The SPiCT main result charts by case are presented in the Annex A, the fit of models by case in Annex B, the retrospective patterns in Annex C and the summary of main results in the Annex D.

The increase in the abundance indices in the last years had important effects on the retrospective patterns (Annex C). The best retrospective pattern was found in the case 7 (free productivity curve,

prior r , and with 6 indices) and the worst in the case 1 (free productivity curve and free r , with one global index) (Figures 6 and 7).

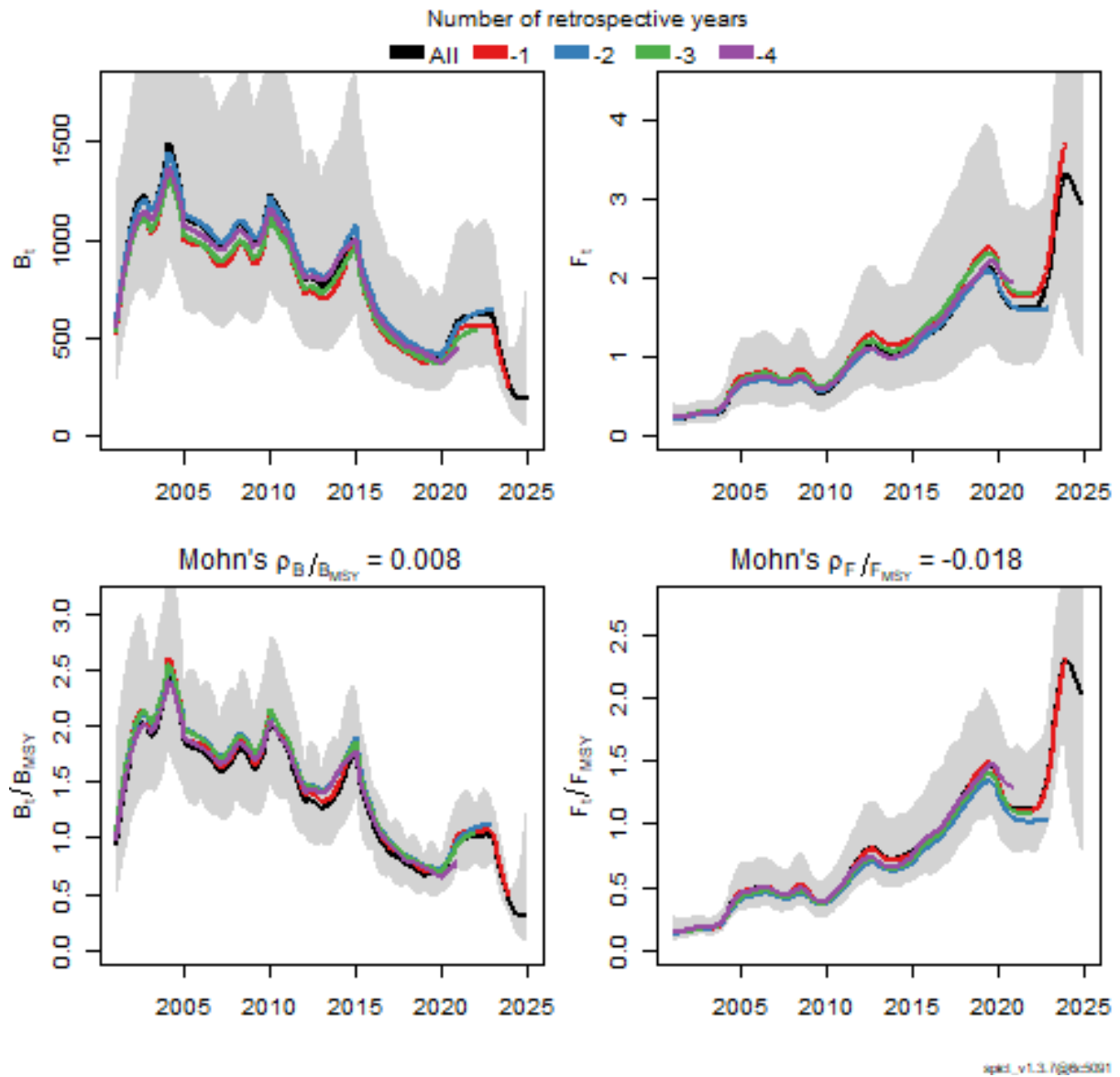


Figure 6. Retrospective patterns of case 7.

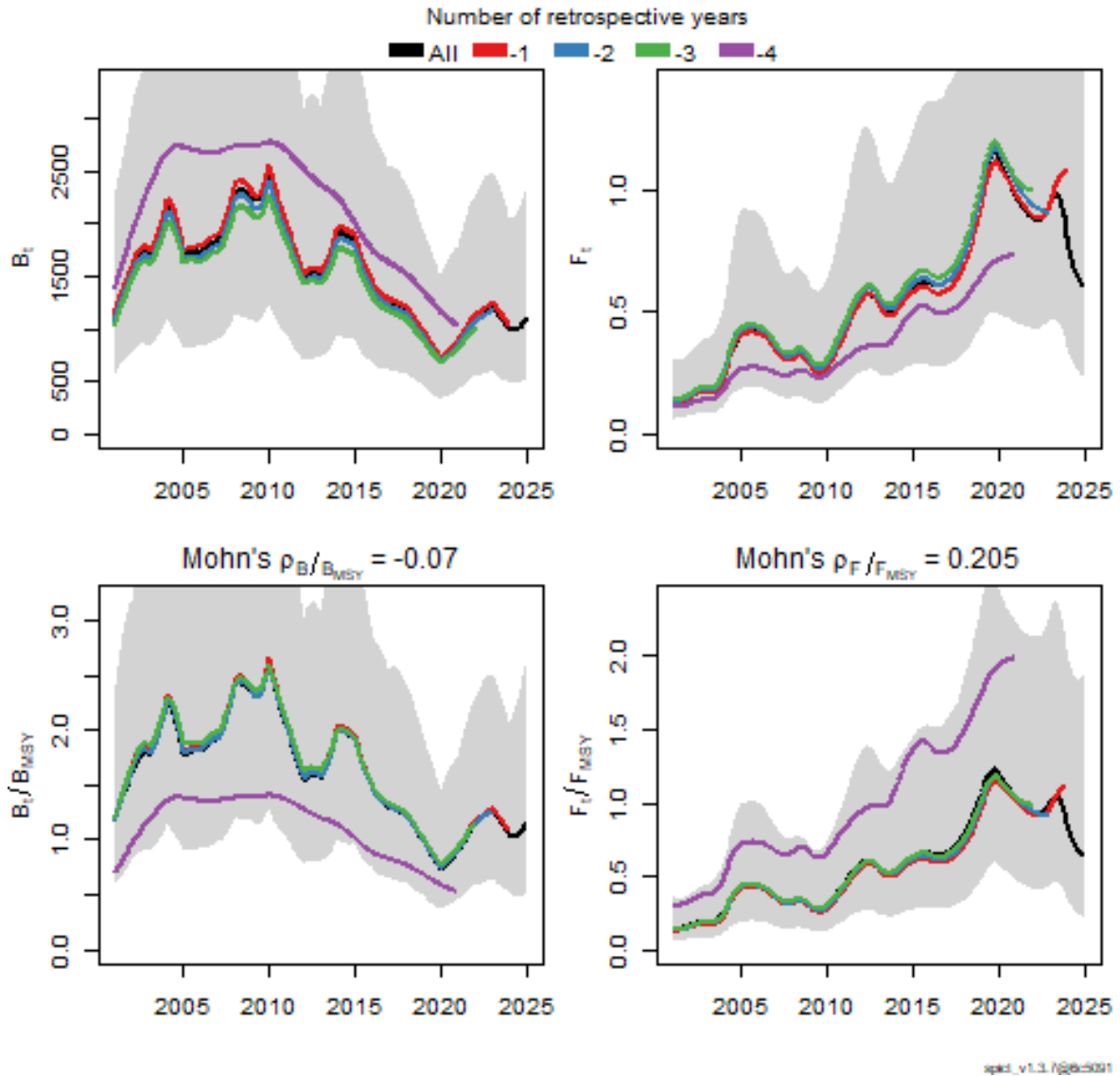


Figure 7. Retrospective patterns of case 1.

The r prior distributions produced r estimations that were in the range expected for a fast-growing and short-life squid (Table 3 and Table 4). The case 2 had no prior information and estimated extremely large r and a very small K , both with very high uncertainty.



Table 3. Main parameters with uncertainty (cilow = Confident interval lower, ciupp= Confident interval upper) estimated by the different cases.

r parameter by case.

	estimate	cilow	ciupp	log.est	
1	0.77	0.19	3.03	-0.27	
2	2.41	1.06	5.48	0.88	
3	0.89	0.38	2.1	-0.12	
4	1.71	0.93	3.16	0.54	
5	1.17	0.63	2.17	0.16	
6	1.68	1.27	2.21	0.52	
7	1.59	1.2	2.1	0.46	

K (kt) parameter by case.

	estimate	cilow	ciupp	log.est	
1	2957	1395	6266	8	
2	1257	633	2493	7	
3	2781	1520	5086	8	
4	1563	939	2602	7	
5	3247	1733	6084	8	
6	2222	1671	2954	8	
7	1639	1125	2387	7	

Bmsyd (kt) by case.

	estimate	cilow	ciupp	log.est	
1	974	335	2829	7	
2	540	252	1158	6	
3	1003	372	2704	7	
4	618	325	1177	6	
5	1623	866	3042	7	
6	1111	835	1477	7	
7	636	345	1172	6	

Fmsyd (kt) by case.

	estimate	cilow	ciupp	log.est	
1	0.95	0.32	2.81	-0.05	
2	1.74	0.81	3.72	0.55	
3	0.92	0.34	2.51	-0.08	
4	1.48	0.78	2.8	0.39	
5	0.59	0.32	1.08	-0.54	
6	0.84	0.64	1.1	-0.18	
7	1.42	0.79	2.57	0.35	

MSYd (kt) by case.

	estimate	cilow	ciupp	log.est	
1	922	762	1116	7	
2	938	836	1054	7	
3	926	771	1113	7	
4	912	816	1021	7	
5	950	768	1174	7	
6	931	812	1068	7	
7	906	819	1003	7	

The lowest r was estimated by case 1, the largest r by case 2, the lowest K by case 2 and the largest K by case 5. B_{msy} was lowest in case 2 and highest in case 6. The lowest maximum sustainable yield was estimated at 906 thousand tons in case 7 and the largest at 950 thousand tons in case 5.

Table 4. Main population parameters estimated by the different cases.

	1	2	3	4	5	6	7
r	0.77	2.41	0.89	1.71	1.17	1.68	1.59
r_c	1.89	3.48	1.85	2.95	1.17	1.68	2.85
r_{old}	3.99	6.2	23.33	10.67	1.17	1.68	13.73
K	2957	1257	2781	1563	3247	2222	1639
B_{msy}	974	540	1003	618	1623	1111	636
F_{msy}	0.95	1.74	0.92	1.48	0.59	0.84	1.42
MSY_d	922	938	926	912	950	931	906

The general biomass trend by year was similar in all cases, with an increasing period (2001-2004 in cases with 6 indices, and 2001-2010 in cases with one global index) and then a decreasing one (Figure 8). The cases with one global index estimated higher biomass levels than the cases with 6 indices, and also estimated a biomass recovery in the biomass during the last 2 years. Fishing mortality by case had the inversed behaviour of the biomass (Figure 9).

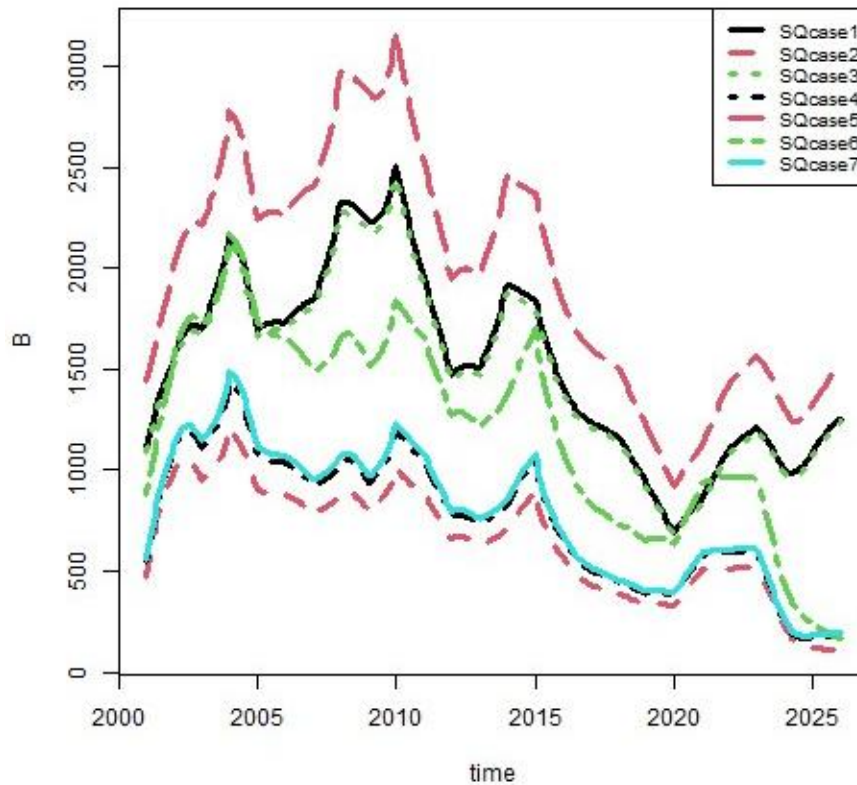


Figure 8. Biomass (thousands of tons) by case

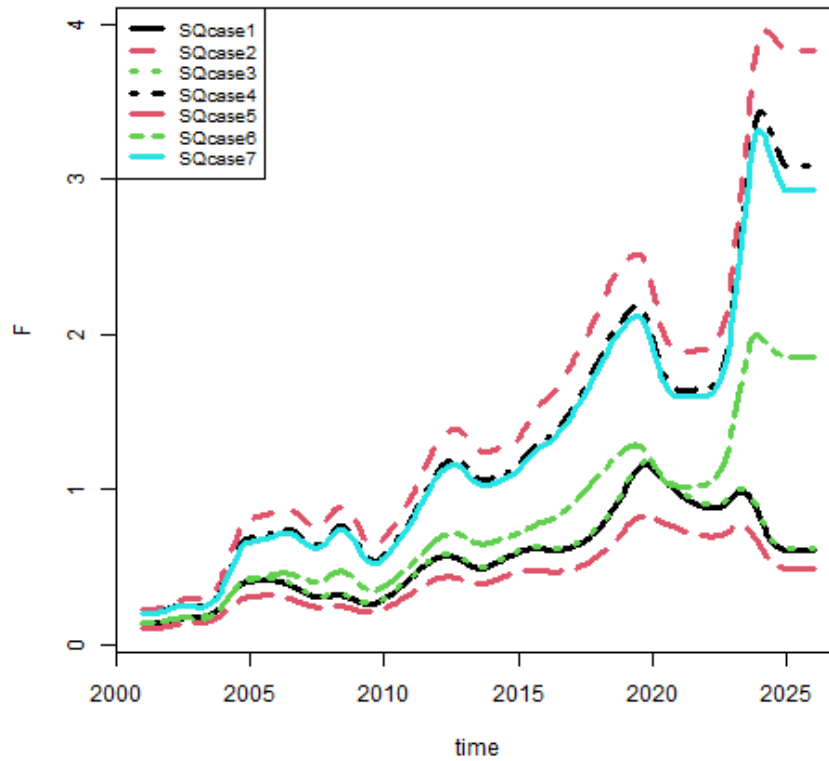


Figure 9. Fishing mortality by case.

All cases had similar patterns in the kobe plot but with different stock status in the year 2025 (Figures 10 and 11). The cases fit to 6 indices (1, 3, and 5) estimated that 2025 stock is in the red zone, while the others in green or in yellow zone. All cases had large uncertainty in the year 2025.

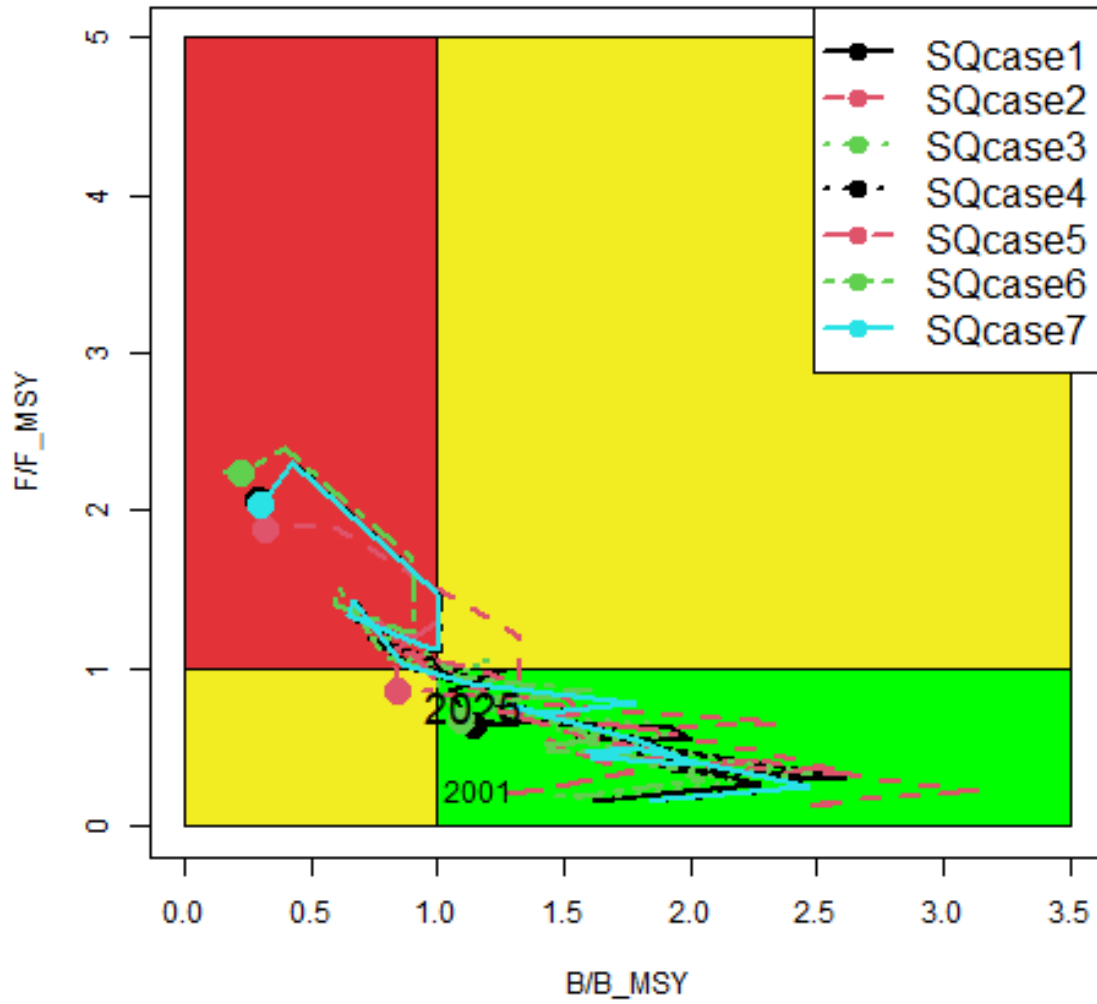


Figure 10. Kobe plot by case.

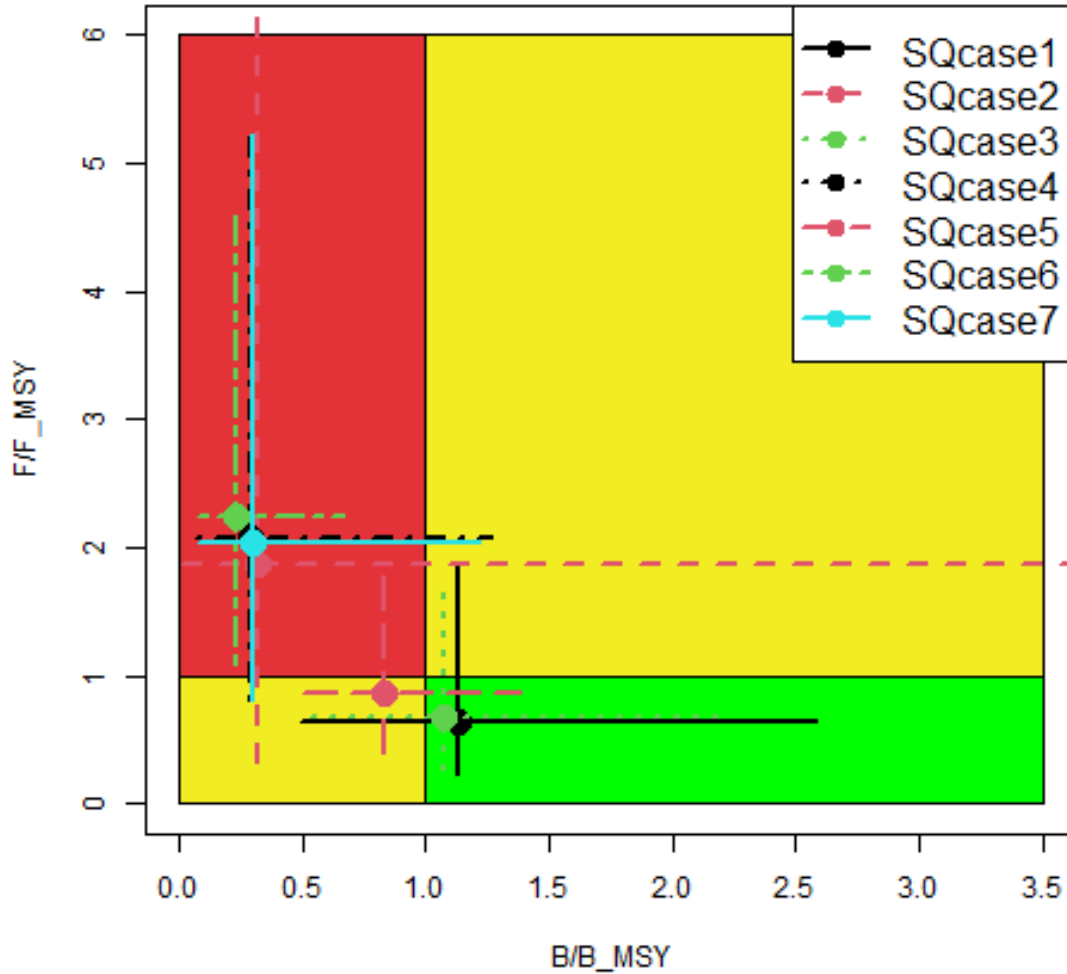


Figure 11. 2025 stock status by case. Lines represent the 95%CI.

The total allowable catches (TAC) for 2026 estimated by case using different harvesting rules are shown in Table 4 and Figure 12. The lowest TAC at Fmsy was estimated at 267 thousand tons by case 6, and the highest at 989 thousand tons by case 1. The current catch estimated by cases with one global index is lower than catch with Fmsy, while the opposite is for cases fit to 6 indices.

Table 4. Total allowable catch (kt) by case and different harvesting rules.

	1	2	3	4	5	6	7
Keep.current.catch	690	471	691	538	688	392	558
Keep.current.F	721	442	726	537	705	375	558
Fish.at.Fmsy	989	457	961	460	794	267	465
No.fishing	1	2	1	2	1	1	2
Reduce.F.by.25.	573	470	578	528	556	344	540
Increase.F.by.25.	852	399	857	520	837	388	549
MSY.hockey.stick.rule	989	385	961	339	794	150	349
ICES.advice.rule	905	0	879	0	717	0	0

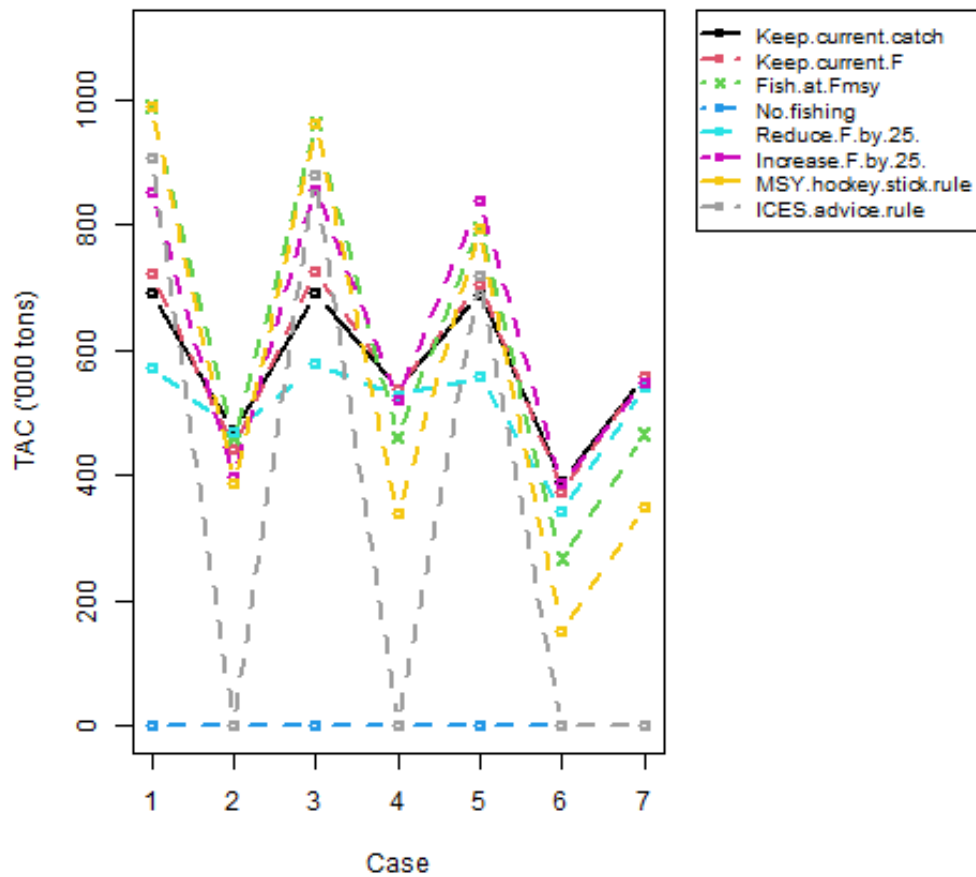


Figure 12. Total allowable catch (TAC) by case and different harvesting rules.

**Sensitivity analysis to shorter time series**

The case 1 and 3 did not converge with the 2012-2024 data series. The estimated parameters by case are presented in table 5.

Table 5. Main parameters with uncertainty (cilow = Confident interval lower, ciupp= Confident interval upper) estimated by the different cases with 2012-2024 time series.

r parameter by case.

	estimate	cilow	ciupp	log.est
2	2.2	0.69	7.04	0.79
4	1.25	0.5	3.17	0.23
5	1.03	0.4	2.69	0.03
6	1.59	1.19	2.12	0.46
7	1.53	1.14	2.04	0.42

K (kt) parameter by case.

	estimate	cilow	ciupp	log.est
2	1378	583	3257	7
4	2035	927	4468	8
5	3663	1327	10114	8
6	2365	1668	3353	8
7	1765	1157	2690	7

Bmsyd (kt) by case.

	estimate	cilow	ciupp	log.est
2	587	253	1358	6
4	740	338	1618	7
5	1831	663	5057	8
6	1182	834	1677	7
7	679	346	1333	7

Fmsyd (kt) by case.

	estimate	cilow	ciupp	log.est
2	1.62	0.7	3.75	0.48
4	1.28	0.61	2.69	0.25
5	0.52	0.2	1.34	-0.66
6	0.8	0.6	1.06	-0.23
7	1.39	0.73	2.67	0.33

MSYd (kt) by case.

	estimate	cilow	ciupp	log.est
2	949	830	1086	7
4	950	790	1143	7
5	944	690	1293	7
6	941	755	1173	7
7	945	808	1106	7

The r parameters range between 1.03 to 2.25 and the K parameter between 17365 kt and 3663 kt. These values were composed each other producing a MSY that ranged between 941 kt and 950 kt (Table 6).

Table 6. Main population parameters estimated by the different cases with 2012-2024 time series.

	2	4	5	6	7
r	2.2	1.25	1.03	1.59	1.53
r_c	3.24	2.57	1.03	1.59	2.78
r_{old}	6.09	54.47	1.03	1.59	15.85
K	1378	2035	3663	2365	1765
B_{msyd}	587	740	1831	1182	679
F_{msyd}	1.62	1.28	0.52	0.8	1.39
MSY_d	949	950	944	941	945

The kobe plot with the short series also showed that the stock is in the red zone, except for the case 5, which is in the yellow one (Figures 13 and 14).

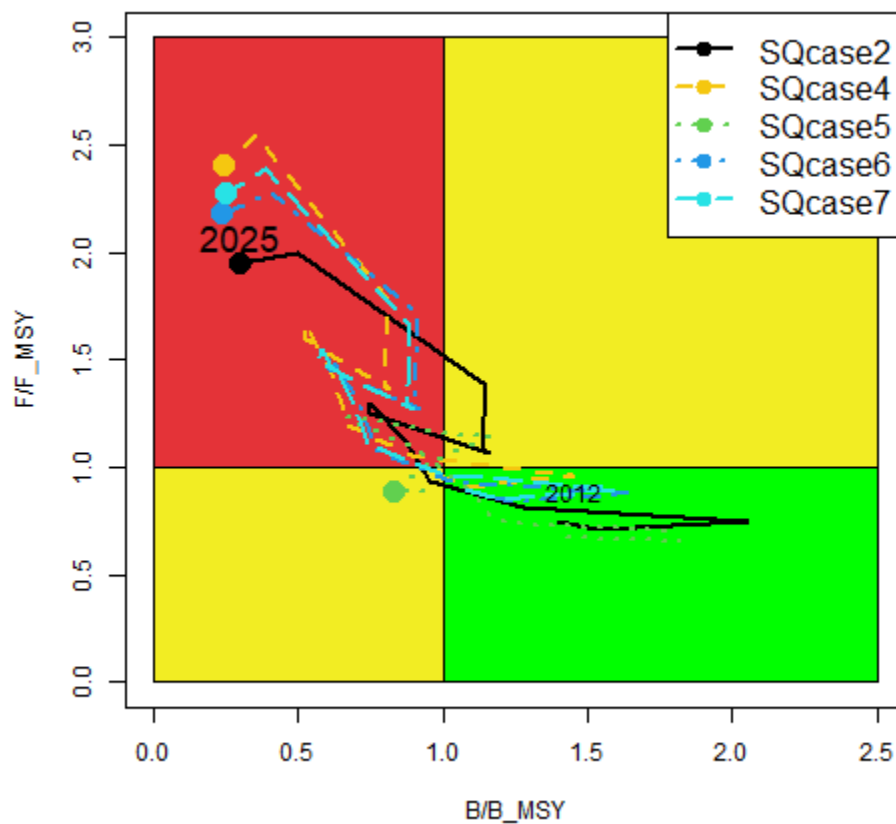


Figure 13. Kobe plot by case with 2012-2024 time series.

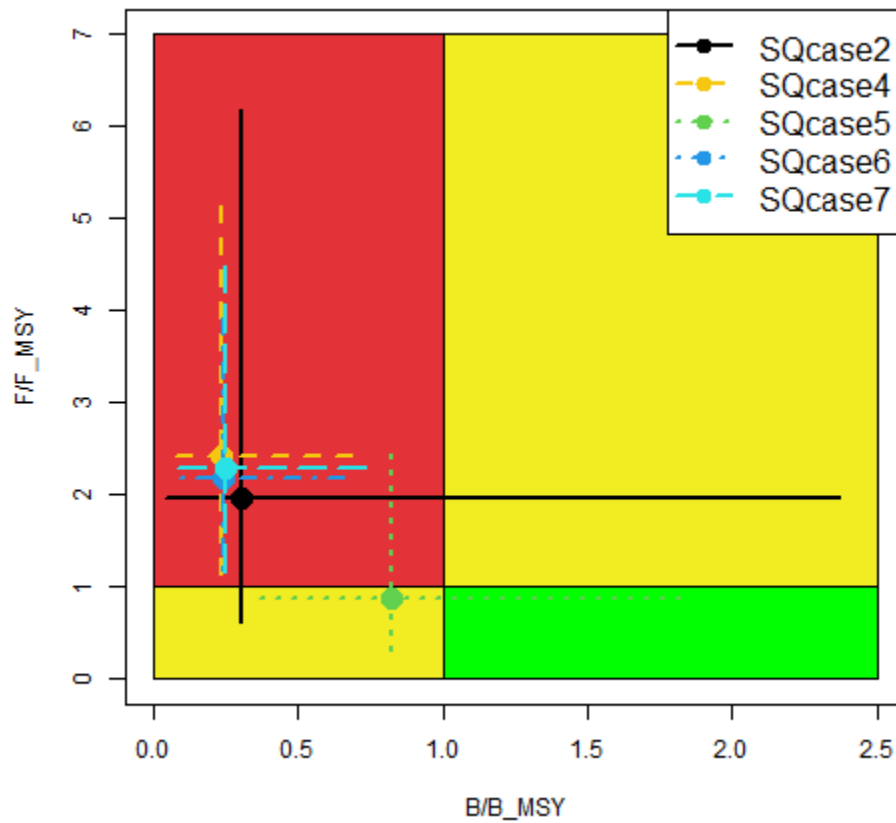


Figure 14. 2025 stock status by case with 2012-2024 time series. Lines represent the 95%CI

The TAC with different harvesting control rules estimated with the short series were similar to ones estimated with the full time series (Tables 4 and 7, and Figures 12 and 15)

Table 7. Total allowable catch (kt) by case and different harvesting rules with 2012-2024 time series.

HCR	2	4	5	6	7
Keep.current.catch	482	534	689	417	515
Keep.current.F	467	534	685	388	513
Fish.at.Fmsy	451	421	756	271	430
No.fishing	2	2	1	1	2
Reduce.F.by.25.	482	521	539	351	509
Increase.F.by.25.	433	522	817	407	494
MSY.hockey.stick.rule	357	261	756	157	282
ICES.advice.rule	0	0	681	0	0

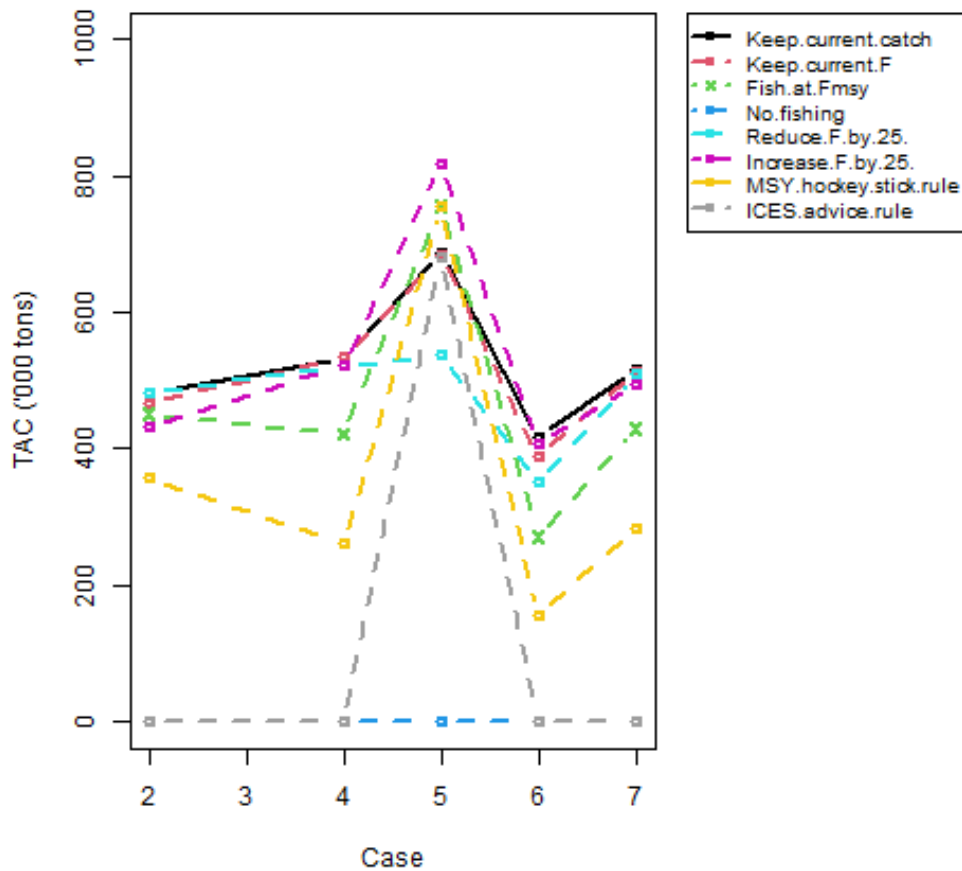


Figure 15. Total allowable catch (TAC) by case and different harvesting rules with short time series.

DISCUSSION

In 2024, the Squid Working Group discussed stock assessment model results that had estimated a healthy squid stock status for the year 2022. Meanwhile, total squid catches in 2024 showed a significant decrease (52%). The current assessment used data up to 2024, representing an improvement in the timeliness of the data used in the model, which reduces the delay between the stock assessment and actual in-season stock conditions.

The updated models estimated population growth parameters (r) that were well within the expected range for fast-growing, short-lived species, as the same informative prior distributions used by Payá (2023 and 2024) were applied. When no prior information was used and the productivity curve was freely estimated, r was large, K was small, uncertainty was higher, and the retrospective pattern was worse. However, estimates of population growth (r) and carrying capacity (K) varied widely across cases and were negatively correlated, resulting in an MSY of approximately 950 kt.



The updated CPUE indices had a greater effect on the retrospective patterns in the cases fitted to a single global index. Case 7 (productivity curve freely estimated with six indices and an r prior) showed the best retrospective pattern, while Case 1 (productivity curve and r parameter freely estimated with one global index) showed the worst. Case 7 estimated that the stock status in 2025 was in the red zone, with a TAC for 2026 of 467 kt, which is similar to the 2024 catch.

The sensitivity analysis, which shortened the data series from 2001–2024 to 2012–2024, showed results similar to those obtained using the full data series. This is likely due to the sharp decline in catch in 2024 and the decreasing trend also present in the shorter data series.

SPiCT is based on the assumption that abundance indices are unbiased indicators of the entire stock. Therefore, the available squid abundance indices need to be improved. Nominal CPUE series should be standardized. The CPUE model should incorporate spatiotemporal variability and account for the presence of different squid sizes across time and regions. For example, the Chinese fleet catches smaller squid when fishing in equatorial waters, but larger squid when operating off the southern coast of Peru.

The main assumption when fitting SPiCT to *D. gigas* in FAO Area 87 is that the stock is a single unit distributed throughout the entire area. However, three phenotypes have been defined based on maturity length or longevity (Nigmatullin et al., 2001). These three phenotypes are present in Peruvian waters, although their relative predominance appears to vary by year depending on environmental conditions. The smaller phenotype is found in equatorial waters, while the larger phenotype is observed in the Chilean EEZ and in international waters south of Peru. It is expected that these three phenotypes have different intrinsic growth rates and carrying capacities, and therefore distinct productivity curves. Although SPiCT, as a state-space model, estimates process error for biomass, it is unclear whether this process error can adequately account for the presence of multiple phenotypes. Similarly, interannual variability in productivity, which also seems to depend on environmental conditions, could potentially be captured as part of the biomass process error.

CONCLUSIONS

1. Squid catch experienced a significant decrease (52%) in 2024.
2. Abundance indices showed a general downward trend.
3. MSY was estimated at approximately 950 kt.
4. The SPiCT model fitted to six indices estimated lower biomass levels and showed better retrospective patterns than the model fitted to a single global index.
5. The 2025 stock status was uncertain but appeared to fall within the red zone of the Kobe plot.
6. The estimated TAC for 2026 under FMSY ranged from 267 kt to 989 kt.
7. The case with the best retrospective pattern estimated the 2025 stock to be in the red zone and a TAC for 2026 of 467 kt, which is similar to the 2024 catch.



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Annex A.

Main results charts by case

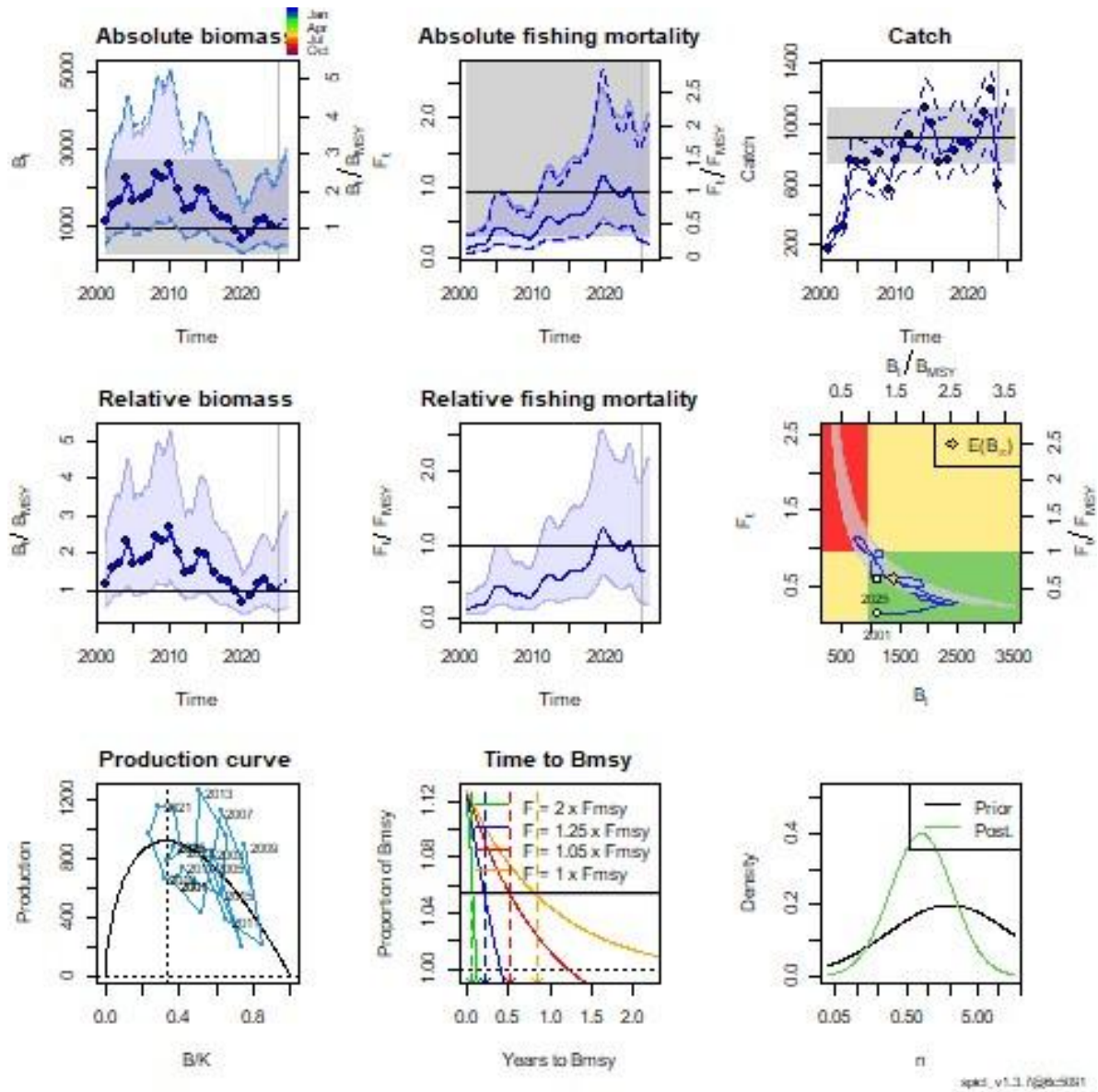


Figure A.1. Main chart results for case 1



Figure A.2. Main results chart for case 2.

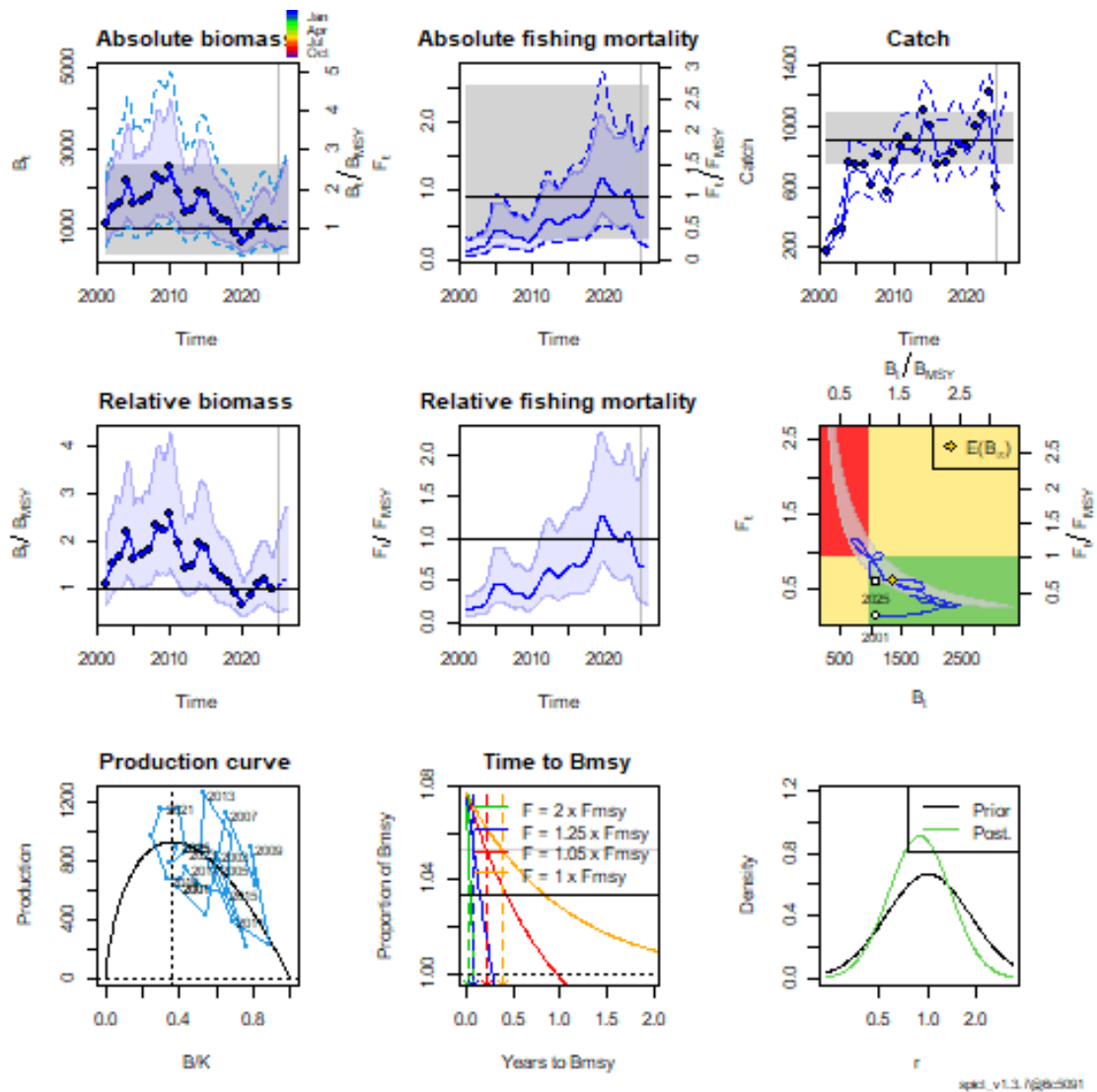


Figure A.3. Main results chart for case 3.

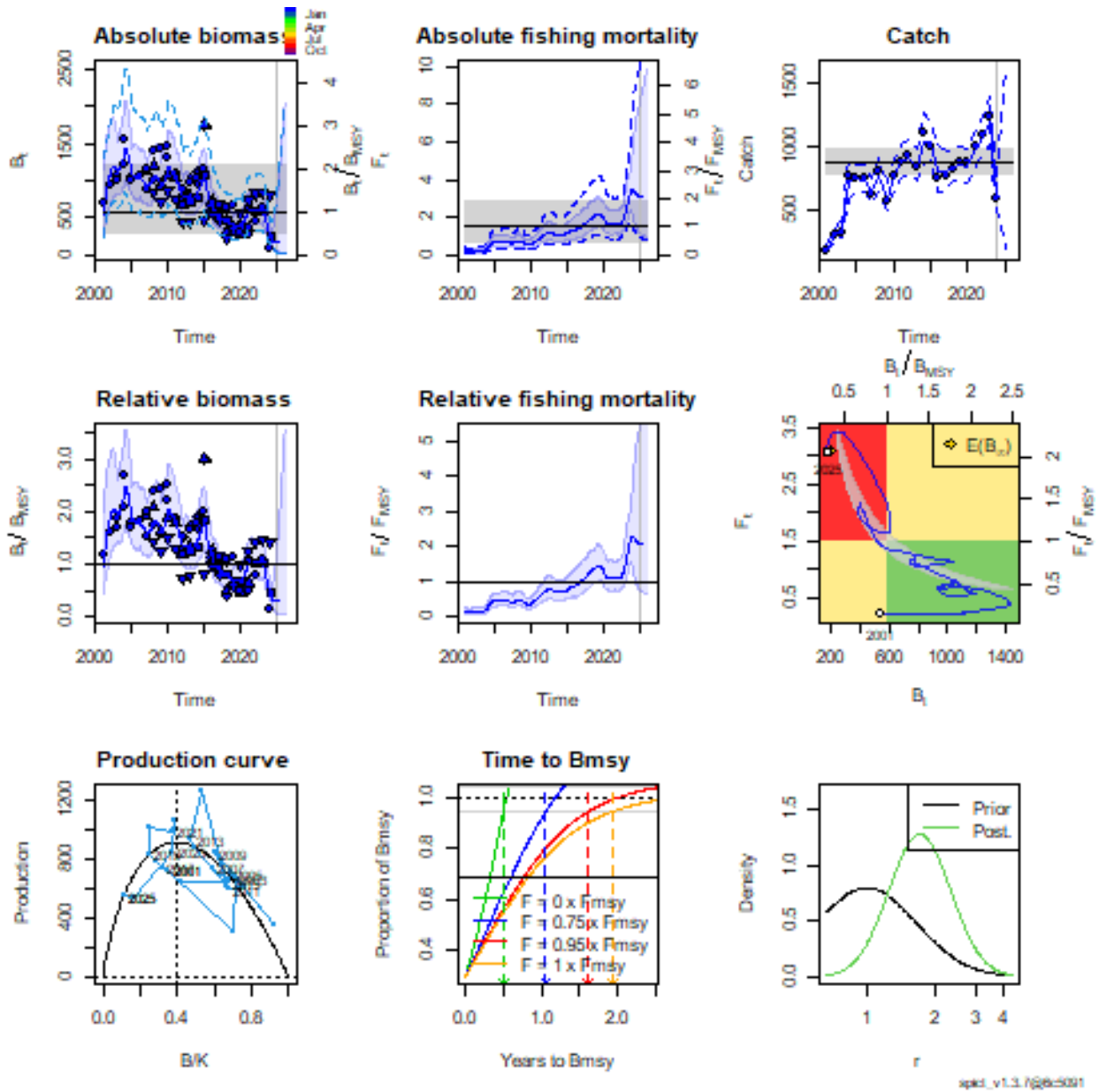


Figure A.4. Main results chart for case 4.

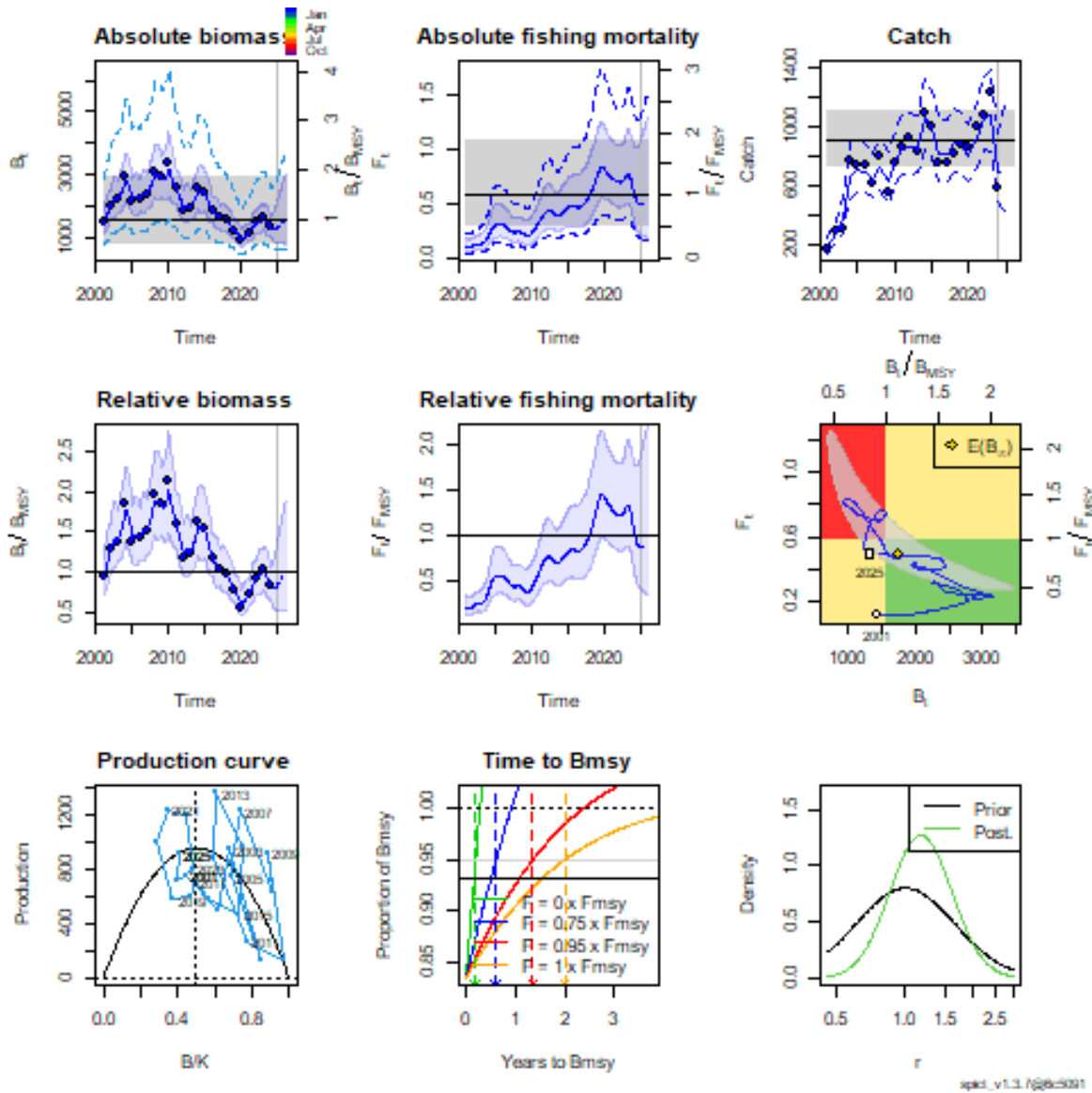


Figure A.5. Main results chart for case 5.

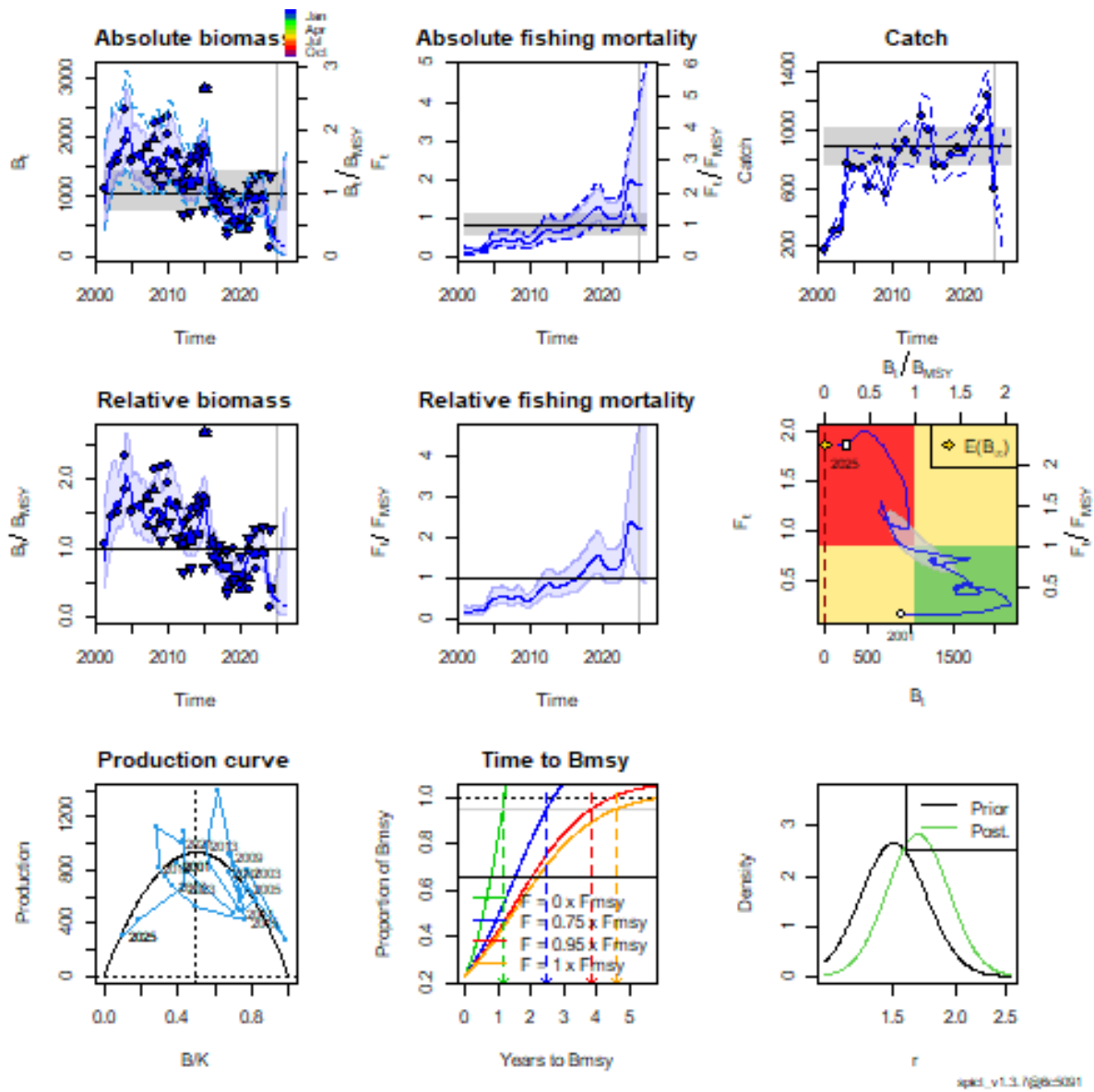


Figure A.6. Main results chart for case 6.

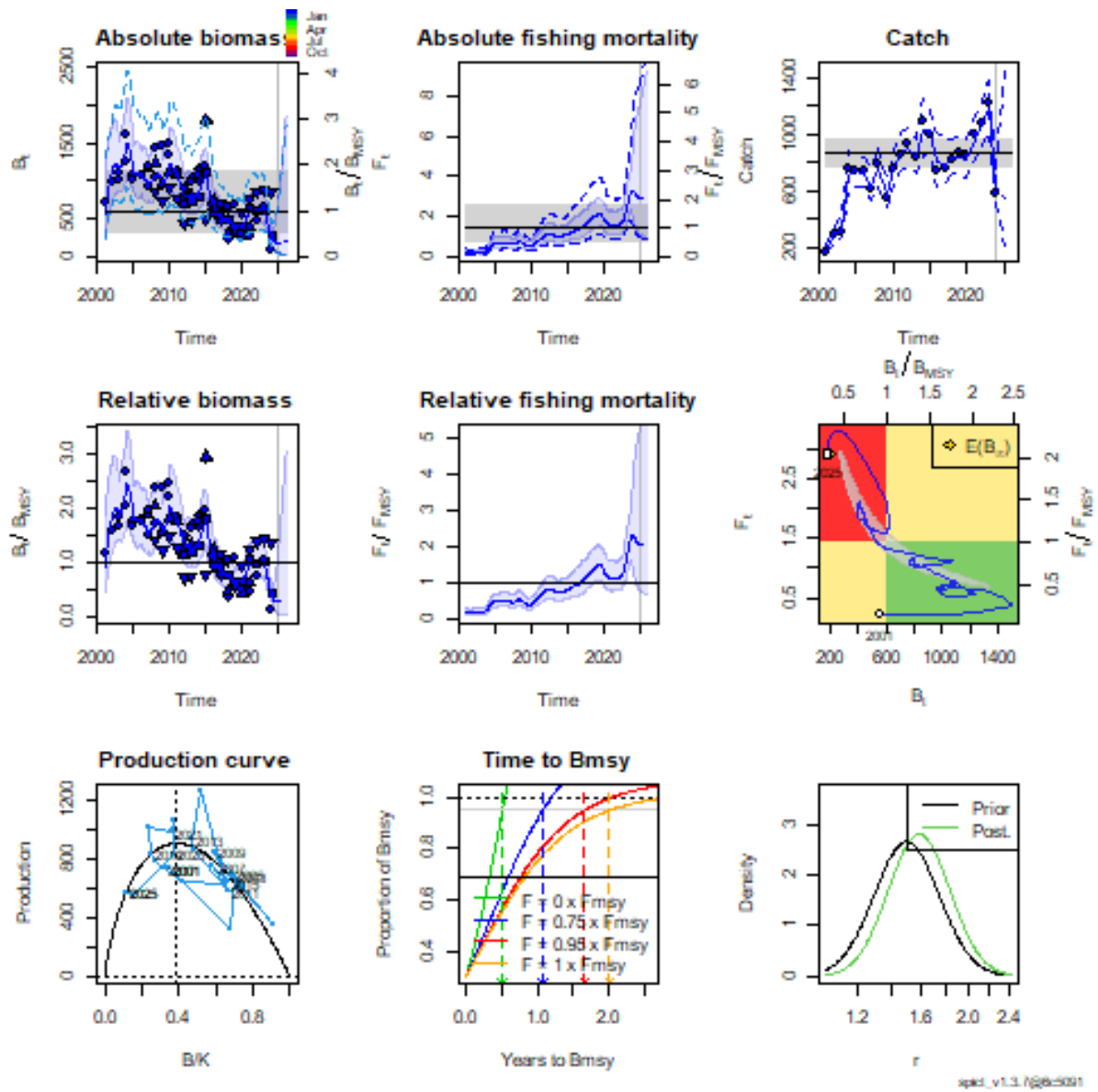


Figure A.7. Main results chart for case 7.



Annex B.

Fit of Models by case

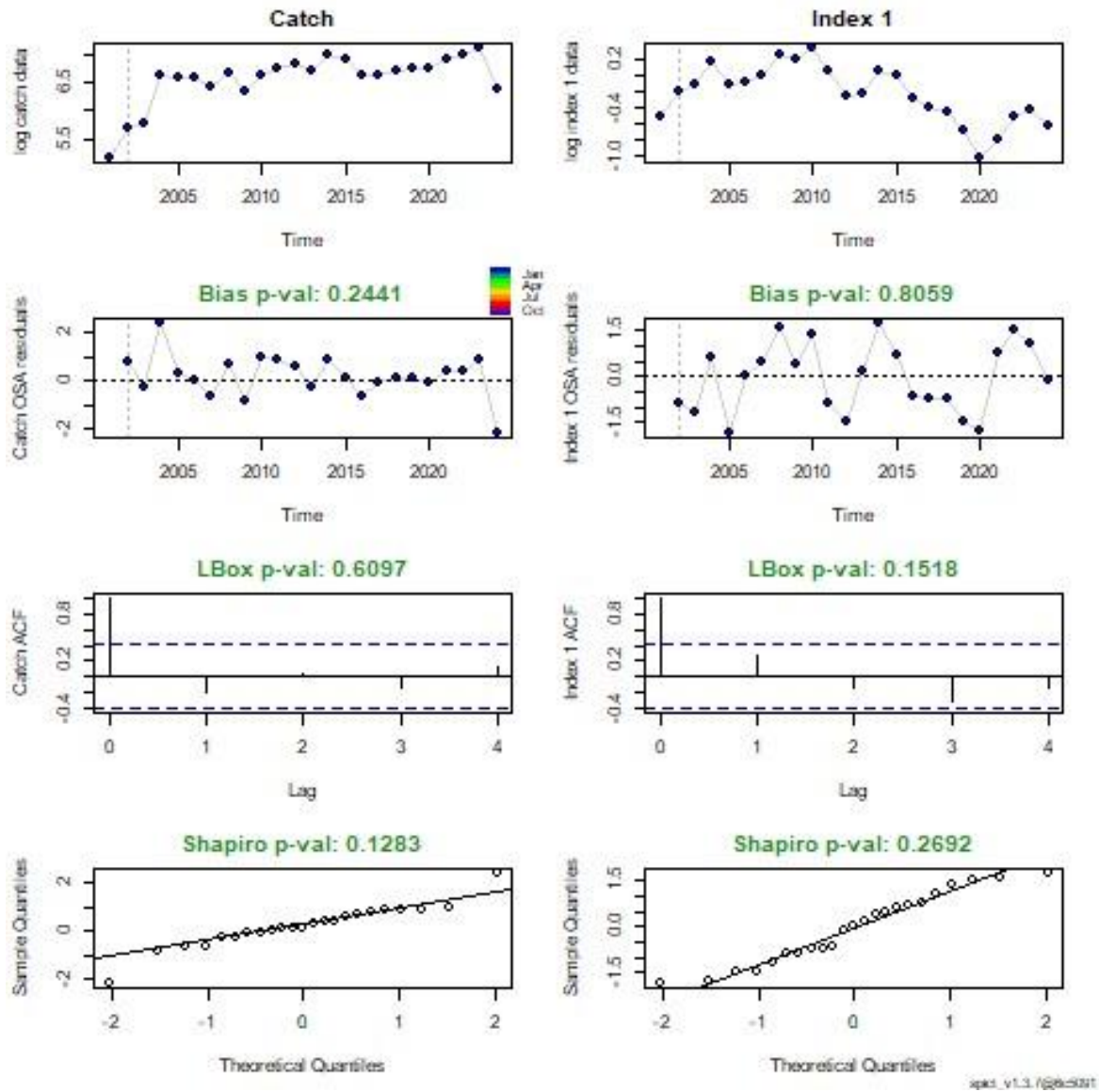
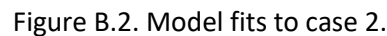


Figure B.1. Model fits to case 1.



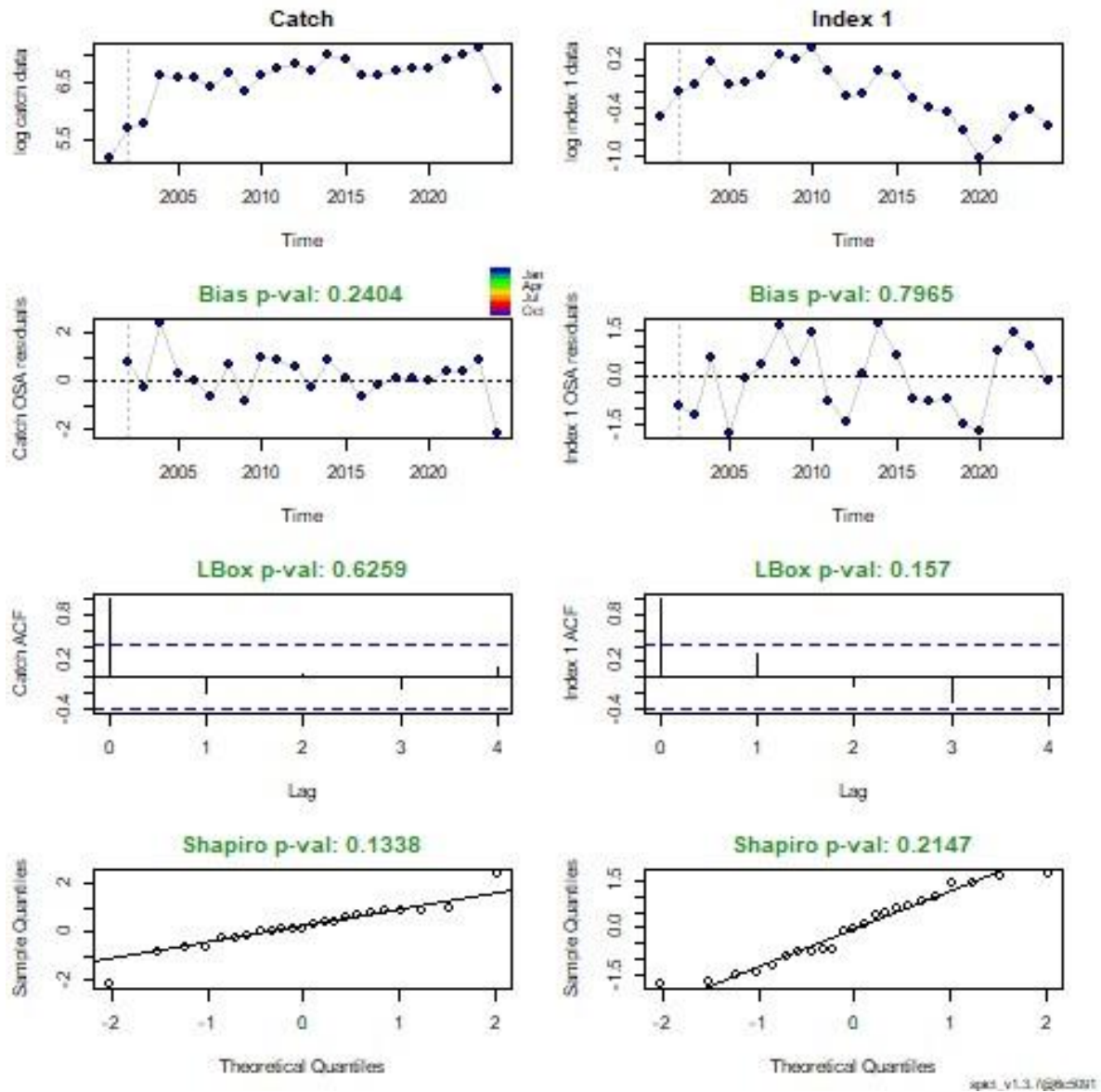


Figure .B3. Model fits to case 3.

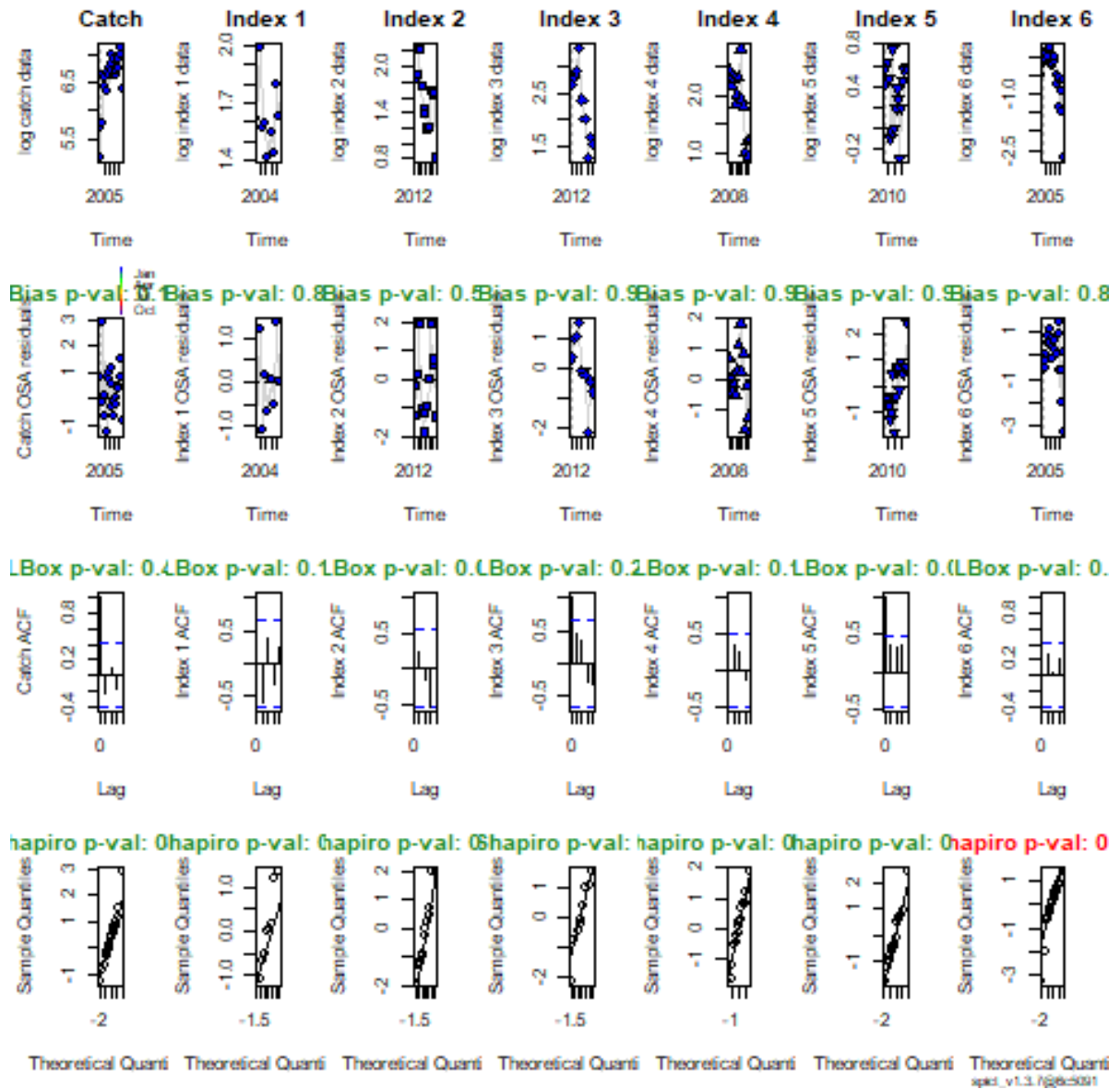


Figure B.4. Model fits to case 4.

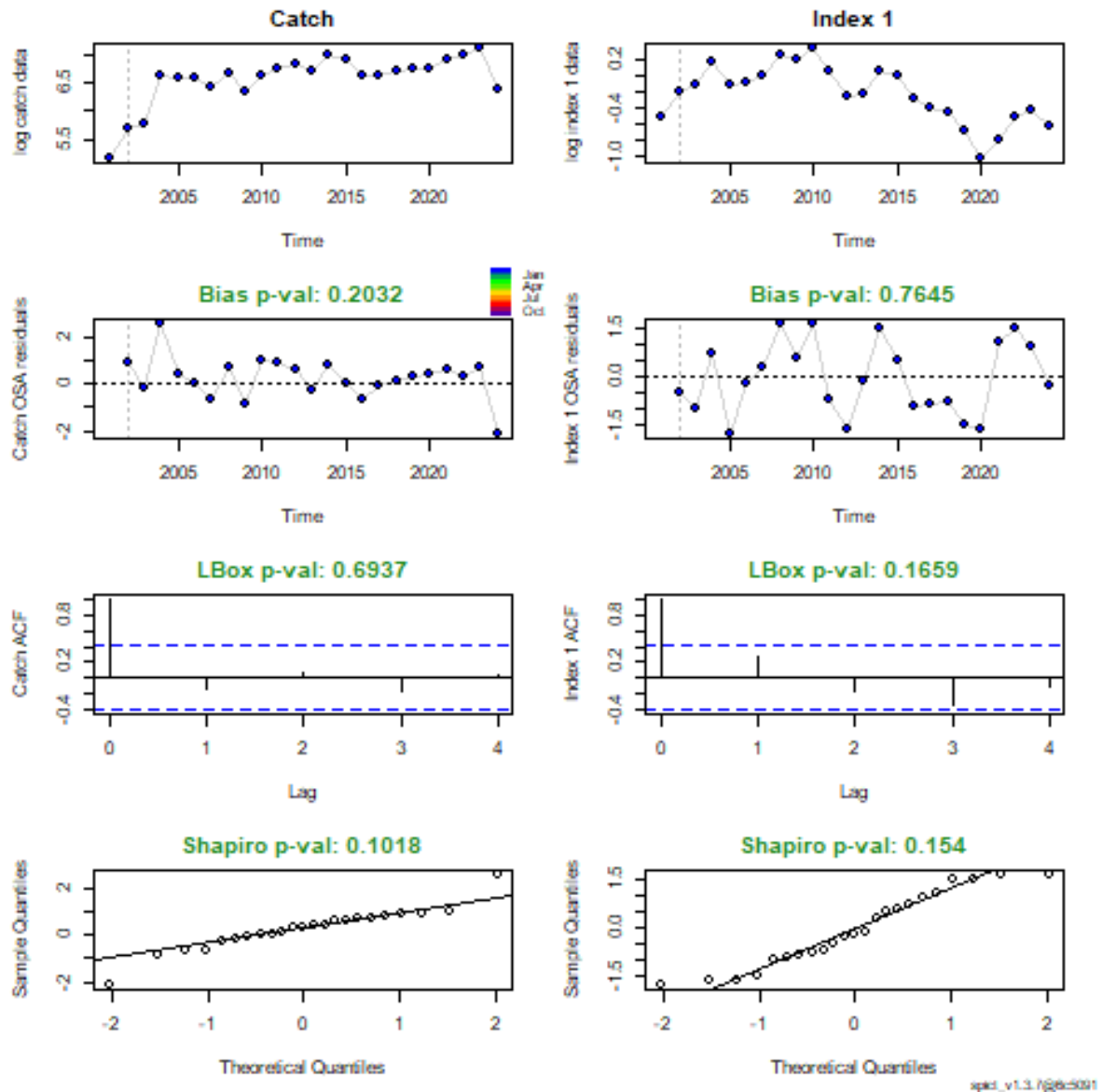


Figure B.5. Model fits to case 5.

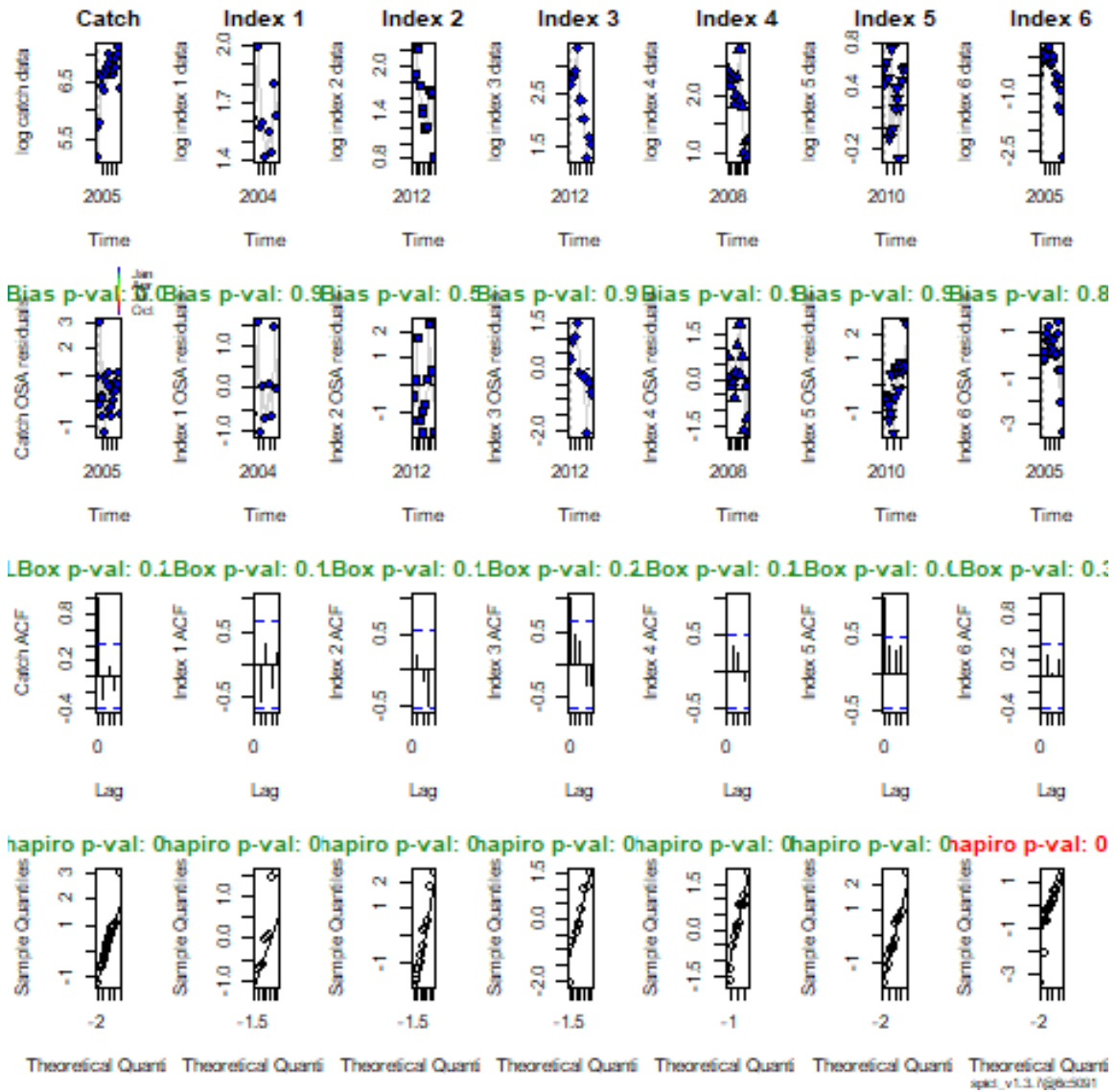


Figure B.6. Model fits to case 6.

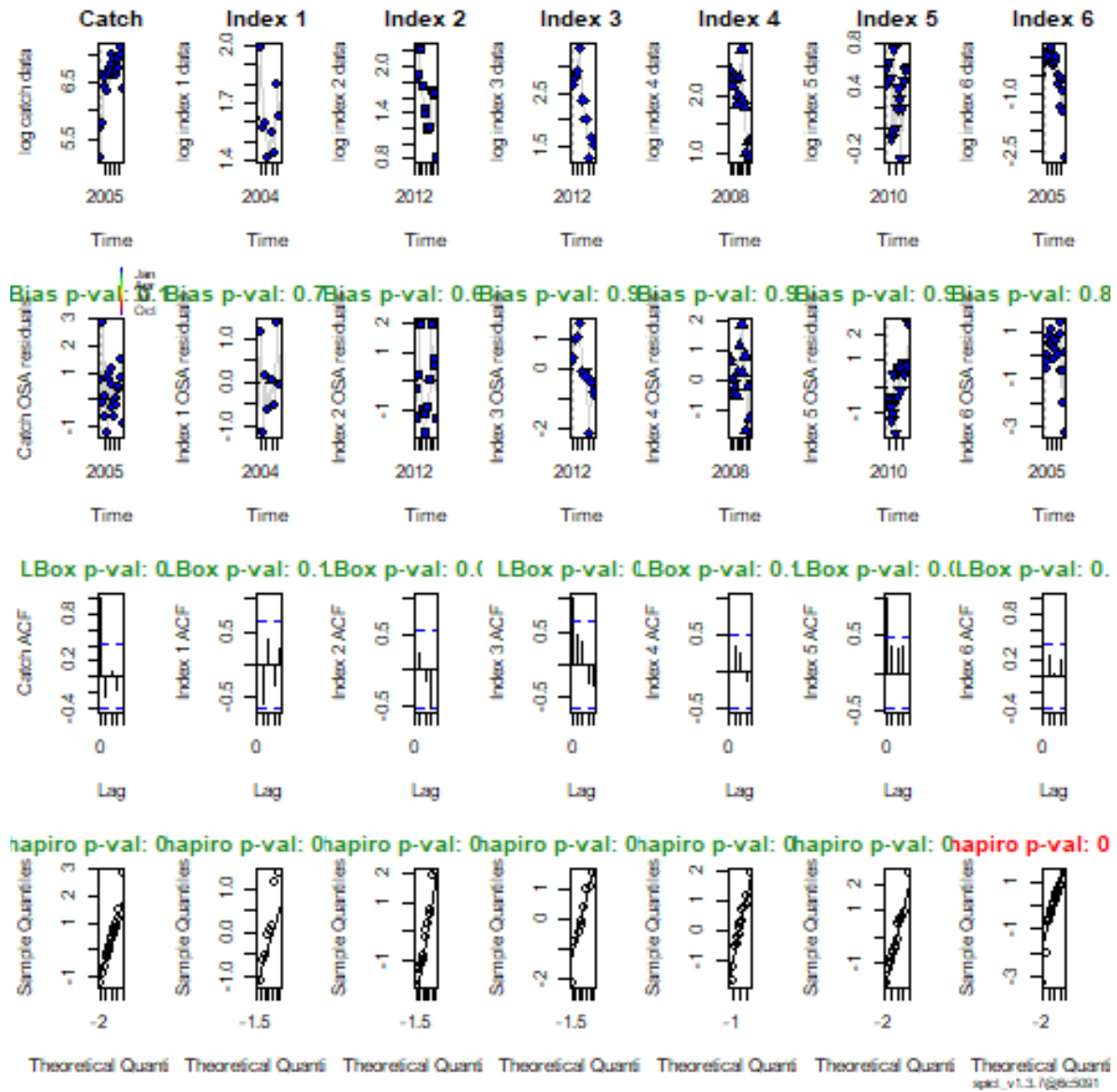


Figure B.7. Model fits to case 7.



Annex c.

Retrospective patterns by case

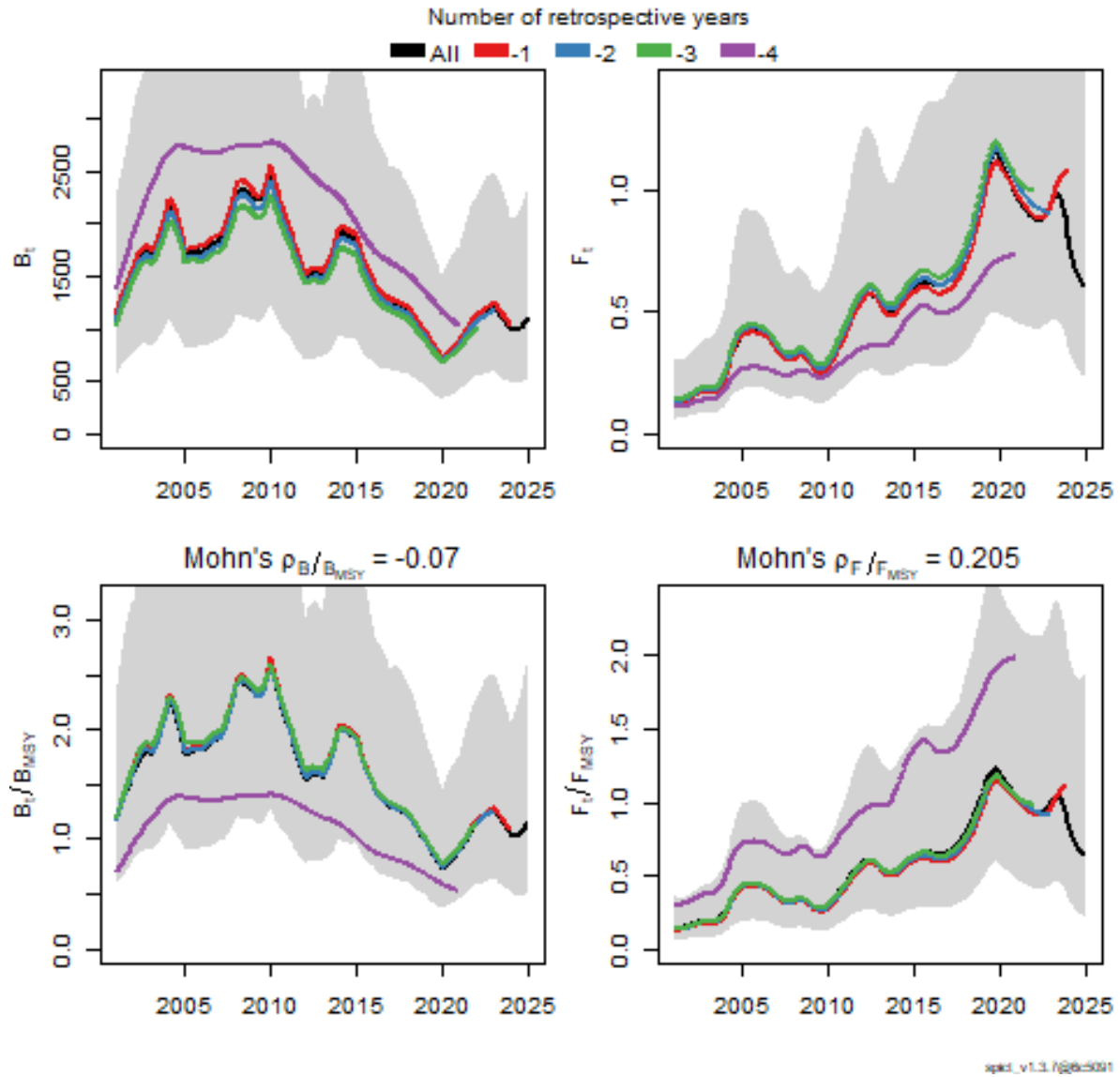


Figure C.1. Retrospective patterns of case 1.

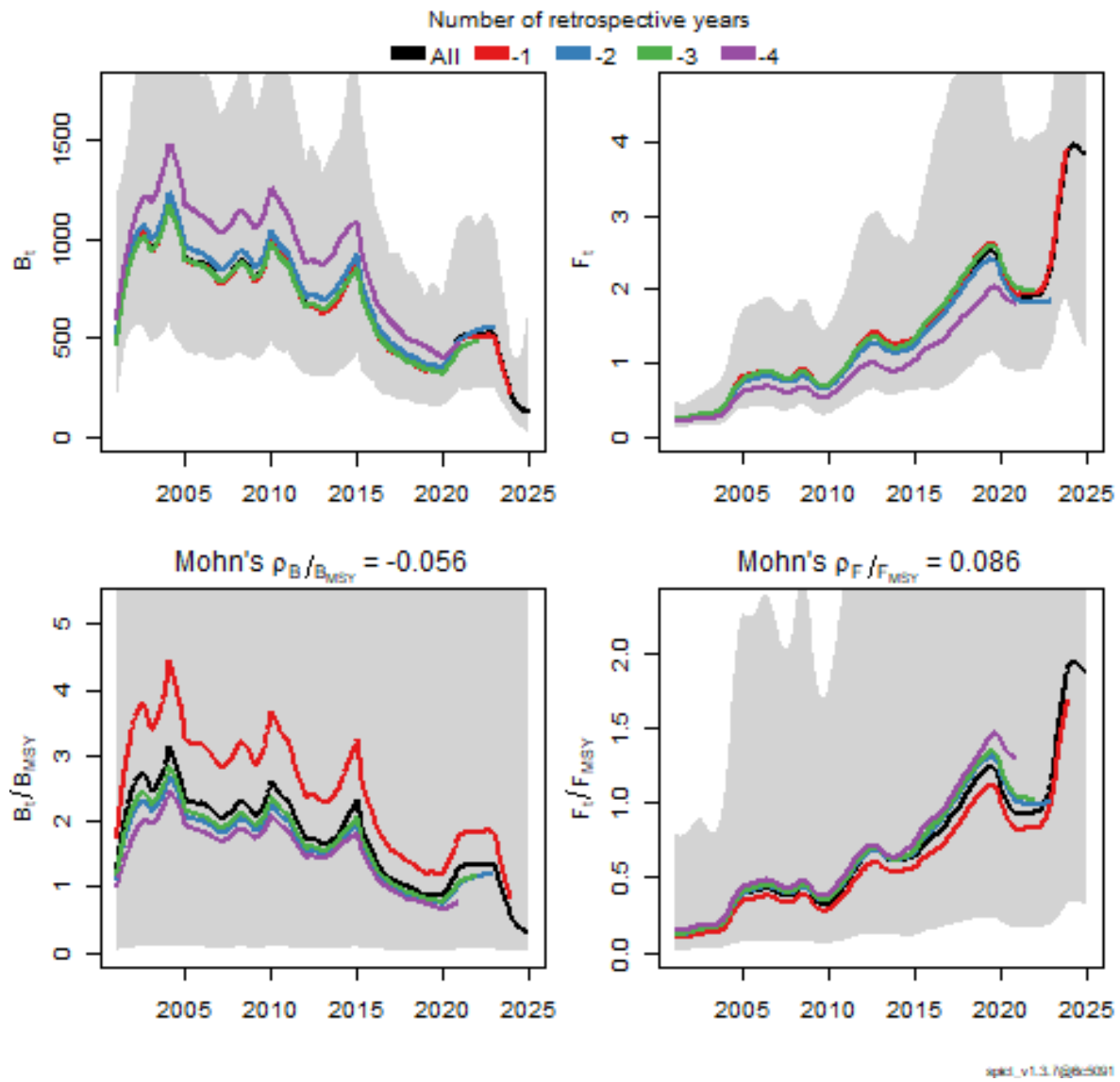


Figure C.2. Retrospective patterns of case 2.

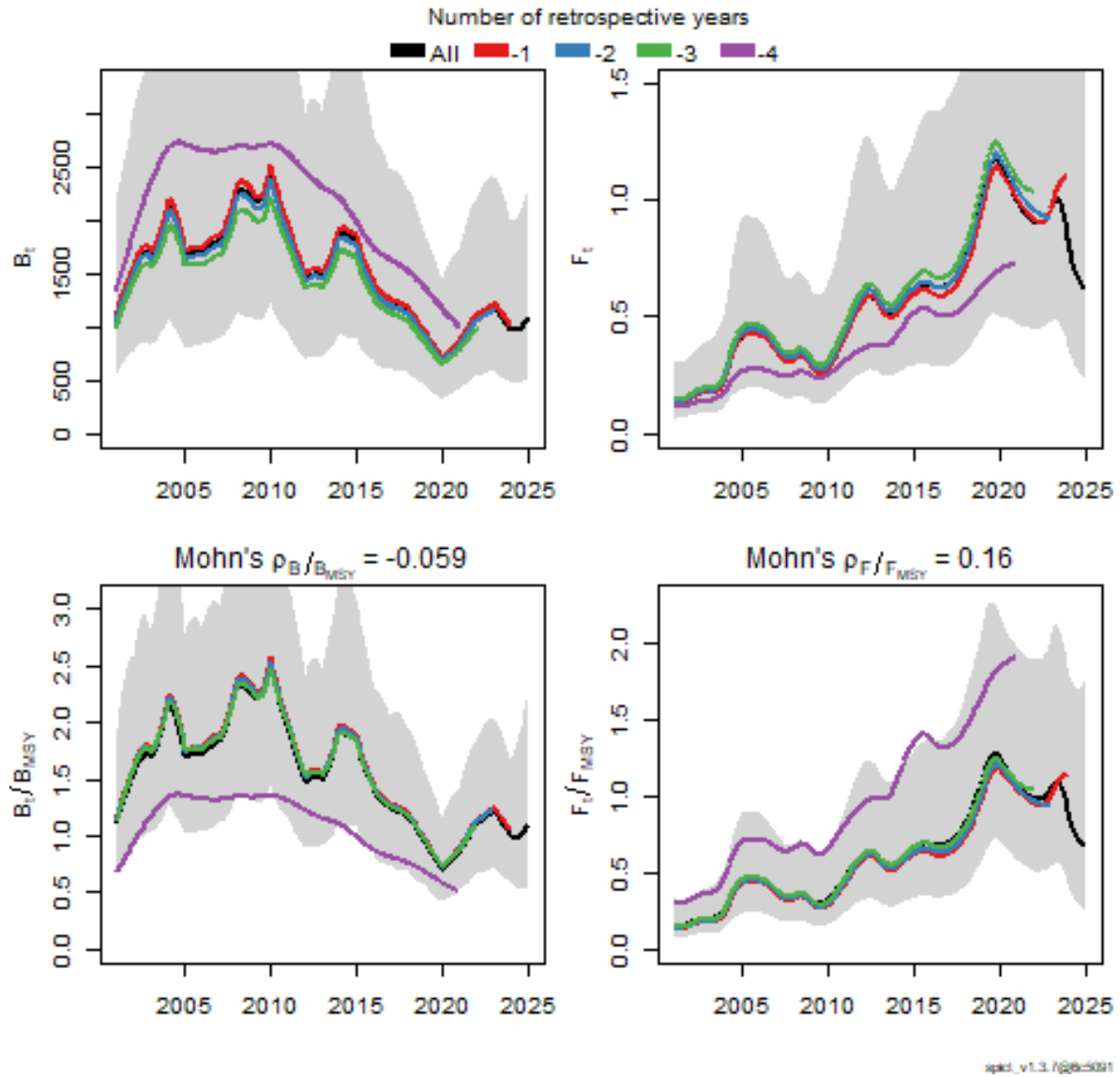


Figure C.3. Retrospective patterns of case 3.

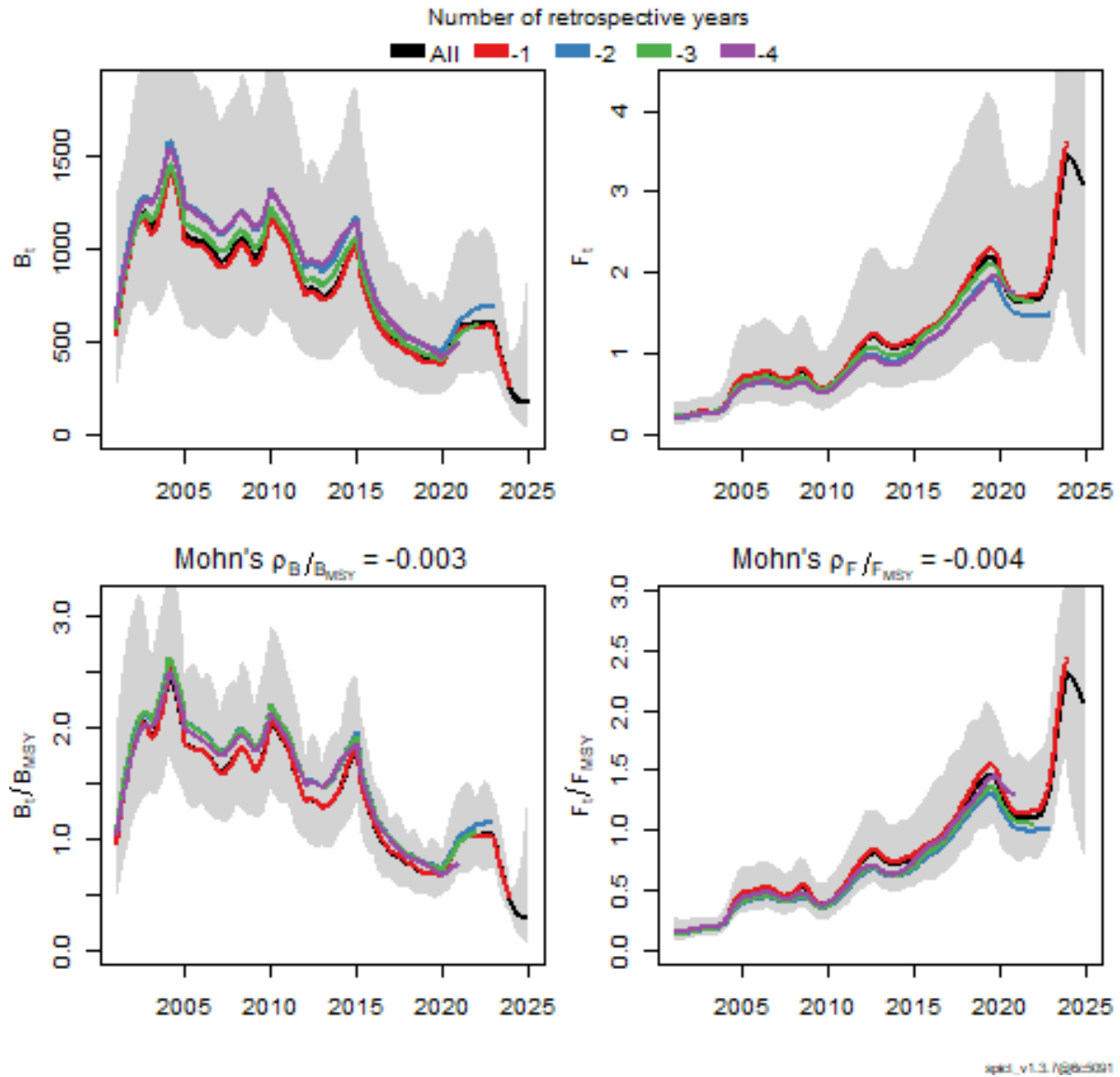


Figure C.4. Retrospective patterns of case 4.

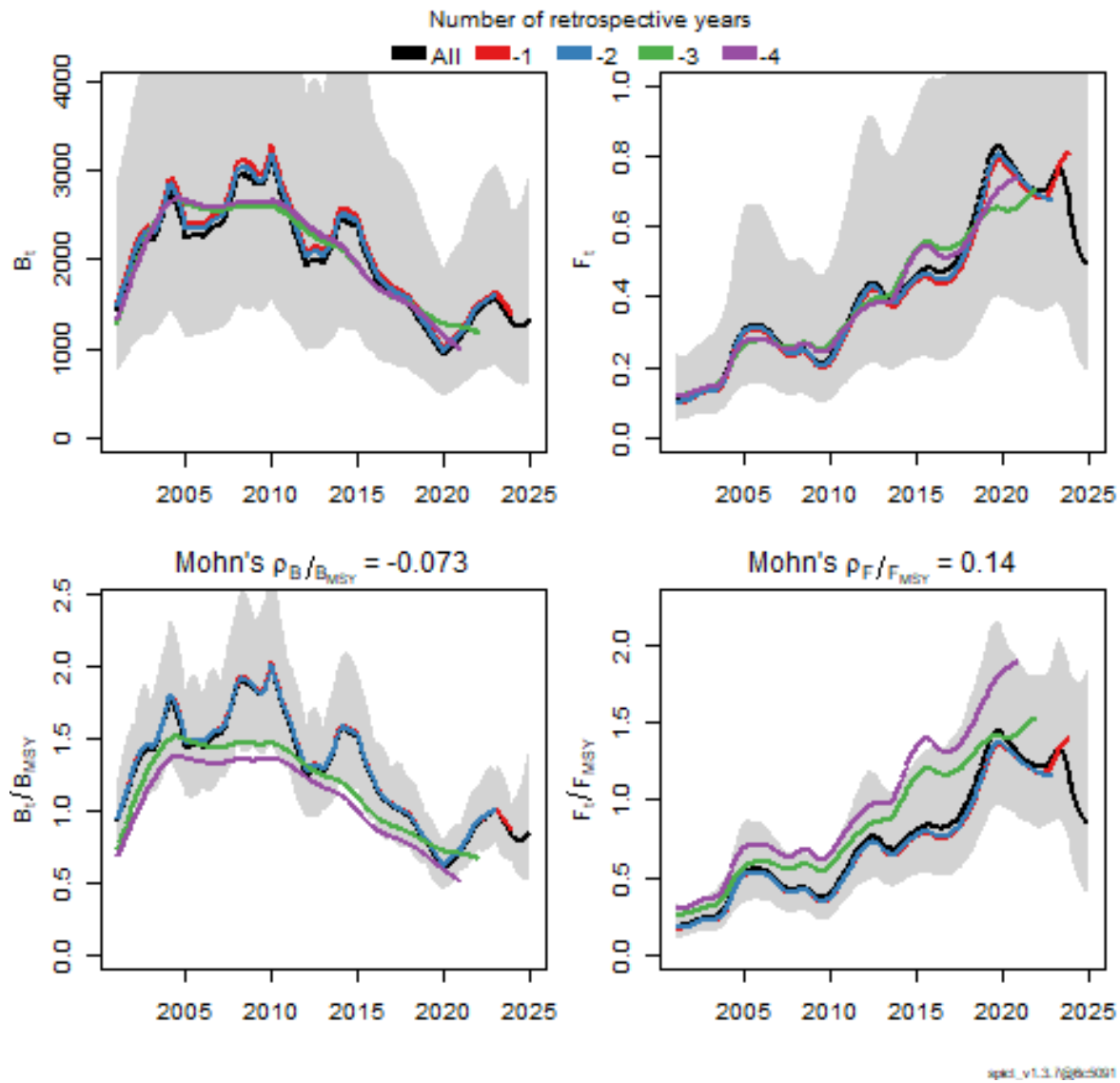


Figure C.5. Retrospective patterns of case 5.

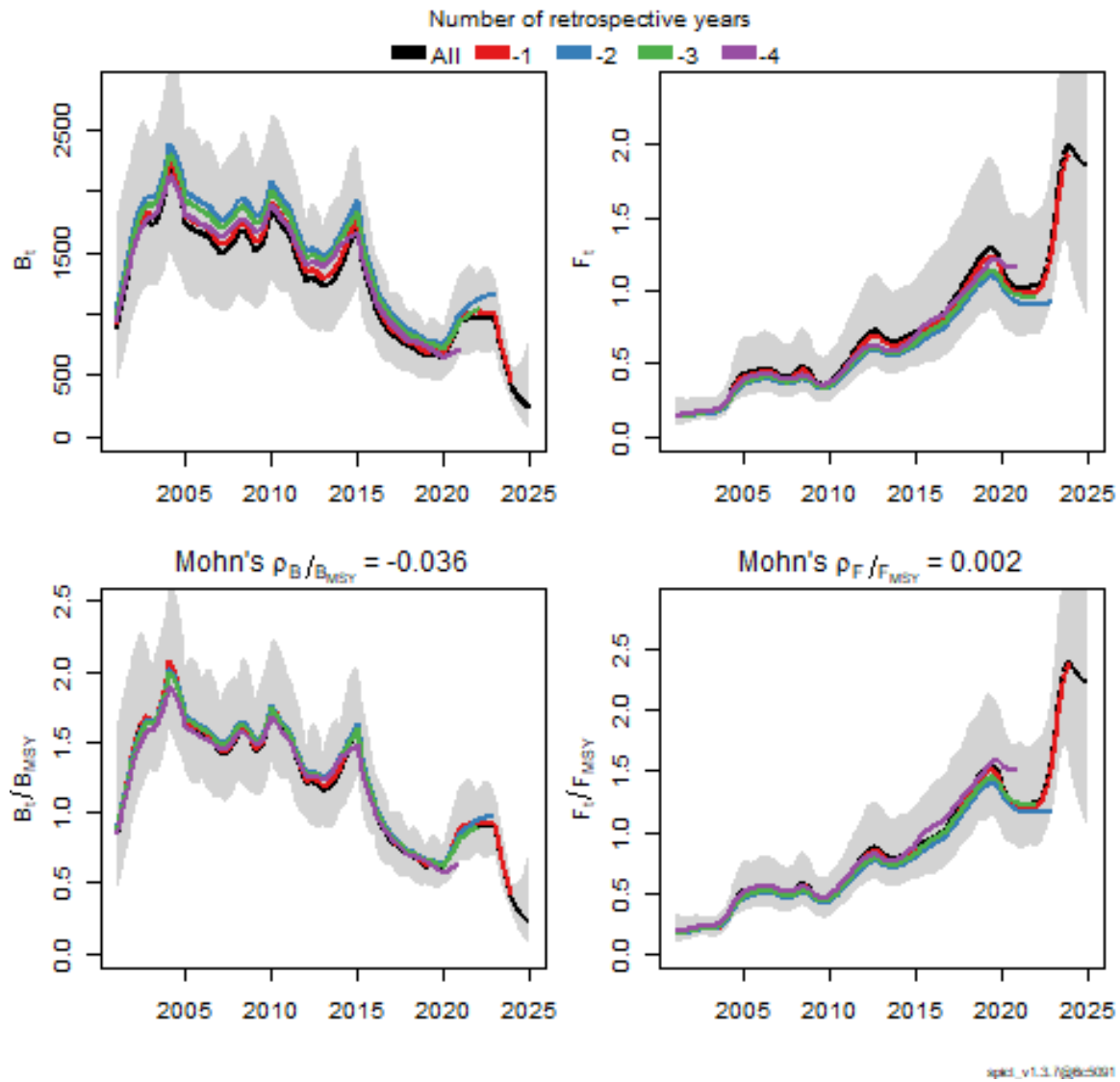


Figure C.6. Retrospective patterns of case 6.

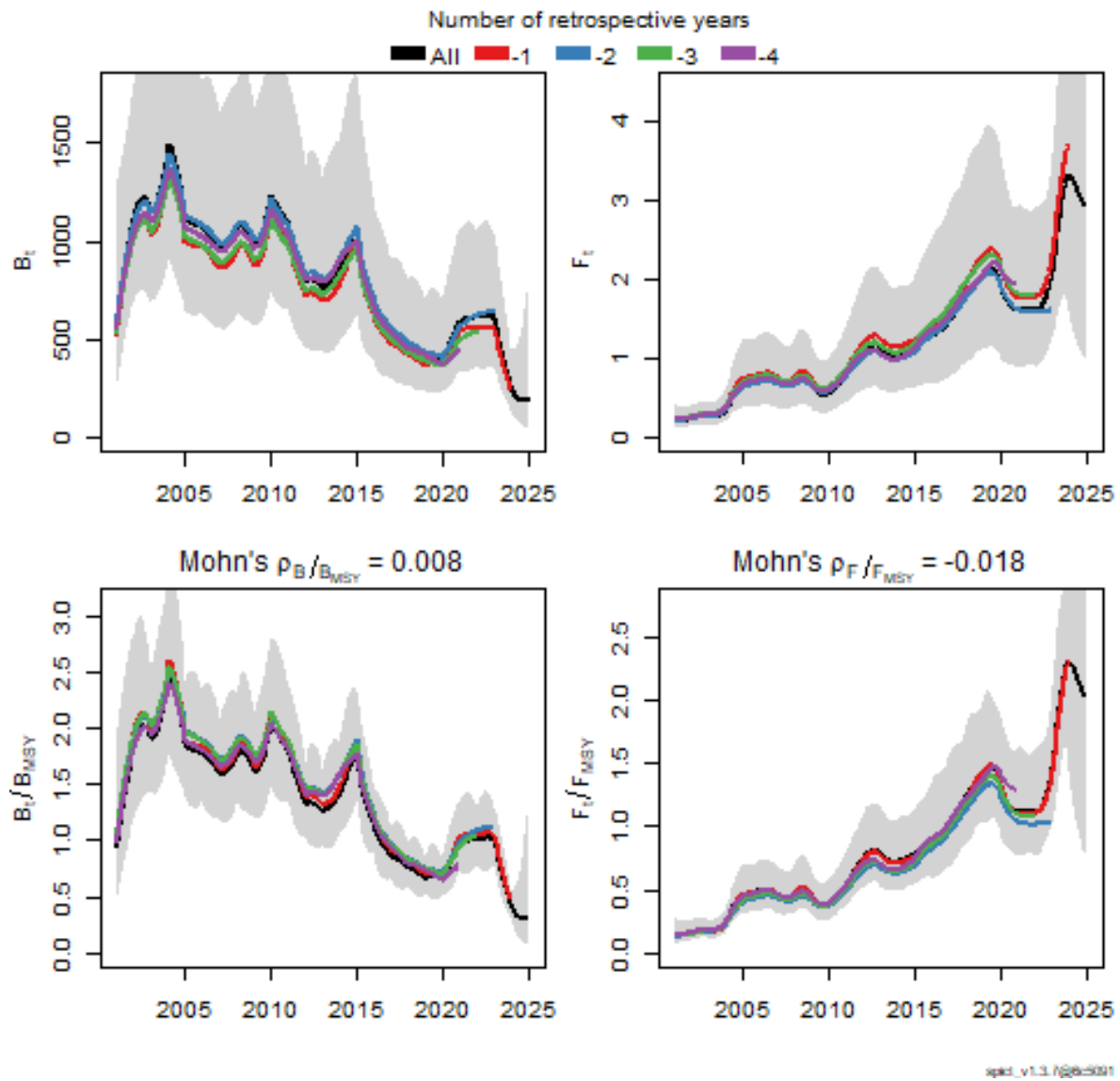


Figure C.7. Retrospective patterns of case 7.



Annex D.

Summary tables by case



Table.D.1. Summary of main results for case 1.

Convergence: 0 MSG: relative convergence (4)
 Objective function at optimum: 11.8511757
 Euler time step (years): 1/16 or 0.0625
 Nobs C: 24, Nobs I1: 24

Priors

logn ~ dnorm[log(2), 2^2]
 logalpha ~ dnorm[log(1), 2^2]
 logbeta ~ dnorm[log(1), 2^2]

Model parameter estimates w 95% CI

	estimate	cilow	ciupp	log.est
alpha	0.2629726	0.0291745	2.370381e+00	-1.3357054
beta	0.4557731	0.1342877	1.546896e+00	-0.7857601
r	0.7650050	0.1928697	3.034342e+00	-0.2678728
rc	1.8930058	0.6373594	5.622371e+00	0.6381659
rold	3.9894658	0.0003209	4.959389e+04	1.3836573
m	922.1314886	761.6513962	1.116425e+03	6.8266878
K	2956.8770075	1395.4366783	6.265509e+03	7.9918889
q	0.0005332	0.0002621	1.084600e-03	-7.5366705
n	0.8082437	0.1144592	5.707345e+00	-0.2128916
sdb	0.2034972	0.1282349	3.229316e-01	-1.5921032
sdf	0.3042108	0.1723240	5.370362e-01	-1.1900343
sdi	0.0535142	0.0074285	3.855117e-01	-2.9278085
sdC	0.1386511	0.0599805	3.205062e-01	-1.9757944

Deterministic reference points (Drp)

	estimate	cilow	ciupp	log.est
Bmsyd	974.2510956	335.4773385	2829.297507	6.8816691
Fmsyd	0.9465029	0.3186797	2.811185	-0.0549812
MSYd	922.1314886	761.6513962	1116.424767	6.8266878

Stochastic reference points (Srp)

	estimate	cilow	ciupp	log.est	rel.diff.Drp
Bmsys	957.5060968	339.1656849	2703.156499	6.8643321	-0.0174881381
Fmsys	0.9468857	0.3203792	2.798535	-0.0545769	0.0004042713
MSYs	906.6552058	749.0341685	1097.444812	6.8097622	-0.0170696453

States w 95% CI (inp\$msytype: s)

	estimate	cilow	ciupp	log.est
B_2024.94	1079.4381410	507.6113816	2295.430604	6.9841959
F_2024.94	0.6112750	0.2351301	1.589150	-0.4922083
B_2024.94/Bmsy	1.1273434	0.4919833	2.583224	0.1198639
F_2024.94/Fmsy	0.6455637	0.2230501	1.868425	-0.4376314

Predictions w 95% CI (inp\$msytype: s)

	prediction	cilow	ciupp	log.est
B_2026.00	1255.1266300	551.0071973	2859.024102	7.1349917
F_2026.00	0.6112753	0.1962783	1.903712	-0.4922079



B_2026.00/Bmsy	1.3108289	0.5477667	3.136869	0.2706597
F_2026.00/Fmsy	0.6455640	0.1891384	2.203428	-0.4376310
Catch_2025.00	721.4638937	428.0898409	1215.889984	6.5812823



Table.D.2. Summary of main results for case 2.

Convergence: 0 MSG: relative convergence (4)
 Objective function at optimum: 47.0344887
 Euler time step (years): 1/16 or 0.0625
 Nobs C: 24, Nobs I1: 9, Nobs I2: 13, Nobs I3: 9, Nobs I4: 15,
 Nobs I5: 17, Nobs I6: 24

Priors

logn ~ dnorm[log(2), 2^2]
 logalpha ~ dnorm[log(1), 2^2]
 logbeta ~ dnorm[log(1), 2^2]

Model parameter estimates w 95% CI

	estimate	cilow	ciupp	log.est
alpha1	0.3471088	0.0805943	1.4949501	-1.0581171
alpha2	0.2491602	0.0474606	1.3080500	-1.3896593
alpha3	1.2972538	0.6471634	2.6003748	0.2602495
alpha4	0.9928357	0.5283573	1.8656368	-0.0071901
alpha5	1.7194958	0.9364444	3.1573320	0.5420311
alpha6	1.4202226	0.7980458	2.5274641	0.3508136
beta	0.2734738	0.0671514	1.1137214	-1.2965495
r	2.4146563	1.0631490	5.4842411	0.8815570
rc	3.4757469	1.6230515	7.4432739	1.2458094
rold	6.2004631	0.7180333	53.5431215	1.8246240
m	938.4751389	835.8375656	1053.7162035	6.8442564
K	1256.5747808	633.3706030	2492.9798957	7.1361449
q1	0.0055022	0.0027103	0.0111700	-5.2025998
q2	0.0100556	0.0048195	0.0209804	-4.5996219
q3	0.0189378	0.0087452	0.0410100	-3.9665928
q4	0.0109492	0.0052473	0.0228469	-4.5144892
q5	0.0024879	0.0011664	0.0053062	-5.9963336
q6	0.0010131	0.0004915	0.0020884	-6.8947395
n	1.3894316	0.6999585	2.7580494	0.3288948
sdb	0.2681935	0.1616000	0.4450976	-1.3160464
sdf	0.3274274	0.2142937	0.5002886	-1.1164890
sdi1	0.0930923	0.0271360	0.3193614	-2.3741635
sdi2	0.0668232	0.0146560	0.3046757	-2.7057057
sdi3	0.3479151	0.2165381	0.5590004	-1.0557969
sdi4	0.2662721	0.1799131	0.3940837	-1.3232365
sdi5	0.4611576	0.3256952	0.6529614	-0.7740153
sdi6	0.3808945	0.2802463	0.5176897	-0.9652328
sdC	0.0895428	0.0258980	0.3095959	-2.4130385

Deterministic reference points (Drp)

	estimate	cilow	ciupp	log.est
Bmsyd	540.013502	251.8332956	1157.966748	6.2915941
Fmsyd	1.737874	0.8115258	3.721637	0.5526622
MSYd	938.475139	835.8375656	1053.716203	6.8442564

Stochastic reference points (Srp)

	estimate	cilow	ciupp	log.est	rel.diff.Drp
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Bmsys	387.30796	6.8082435	22033.20976	5.9592201	-0.3942742
Fmsys	2.03868	0.1937689	21.44935	0.7123027	0.1475498
MSys	835.53200	482.5578463	1446.69437	6.7280686	-0.1232067

States w 95% CI (inp\$msytype: s)

	estimate	cilow	ciupp	log.est
B_2024.94	123.8412254	25.6941307	596.89309	4.8190003
F_2024.94	3.8254204	1.2441248	11.76236	1.3416684
B_2024.94/Bmsy	0.3197487	0.0087028	11.74790	-1.1402198
F_2024.94/Fmsy	1.8764199	0.3173502	11.09485	0.6293657

Predictions w 95% CI (inp\$msytype: s)

	prediction	cilow	ciupp	log.est
B_2026.00	109.9370045	5.9294446	2038.32667	4.6999075
F_2026.00	3.8254211	1.0388568	14.08649	1.3416685
B_2026.00/Bmsy	0.2838491	0.0033213	24.25857	-1.2593126
F_2026.00/Fmsy	1.8764203	0.2817116	12.49843	0.6293658
Catch_2025.00	441.7876796	109.9142387	1775.71492	6.0908294
E(B_inf)	177.9688731	NA	NA	5.1816087



Table.D.3. Summary of main results for case 3.

Convergence: 0 MSG: relative convergence (4)
 Objective function at optimum: 12.3027131
 Euler time step (years): 1/16 or 0.0625
 Nobs C: 24, Nobs I1: 24

Priors

logr ~ dnorm[log(1), 0.6^2]
 logn ~ dnorm[log(2), 2^2]
 logalpha ~ dnorm[log(1), 2^2]
 logbeta ~ dnorm[log(1), 2^2]

Model parameter estimates w 95% CI

	estimate	cilow	ciupp	log.est
alpha	0.2583118	0.0291994	2.285148e+00	-1.3535880
beta	0.4562081	0.1340908	1.552126e+00	-0.7848061
r	0.8881754	0.3752362	2.102291e+00	-0.1185860
rc	1.8466421	0.6786779	5.024603e+00	0.6133689
rold	23.3334932	0.0000000	8.942533e+16	3.1498898
m	926.2508982	770.9298848	1.112865e+03	6.8311451
K	2780.6856407	1520.2168964	5.086256e+03	7.9304528
q	0.0005447	0.0002709	1.095100e-03	-7.5153499
n	0.9619356	0.2382227	3.884266e+00	-0.0388078
sdb	0.2073249	0.1324261	3.245857e-01	-1.5734680
sdf	0.3033565	0.1716037	5.362657e-01	-1.1928467
sdi	0.0535545	0.0074923	3.828016e-01	-2.9270561
sdC	0.1383937	0.0598579	3.199713e-01	-1.9776528

Deterministic reference points (Drp)

	estimate	cilow	ciupp	log.est
Bmsyd	1003.173168	372.2366532	2703.539255	6.9109234
Fmsyd	0.923321	0.3393389	2.512302	-0.0797783
MSYd	926.250898	770.9298848	1112.864793	6.8311451

Stochastic reference points (Srp)

	estimate	cilow	ciupp	log.est	rel.diff.Drp
Bmsys	982.1935807	371.3956287	2597.510998	6.8897884	-0.0213599307
Fmsys	0.9234293	0.3387093	2.517562	-0.0796611	0.0001172306
MSYs	906.9885721	755.6014081	1088.706640	6.8101299	-0.0212376724

States w 95% CI (inp\$msytype: s)

	estimate	cilow	ciupp	log.est
B_2024.94	1057.7714886	499.7201822	2239.014076	6.9639196
F_2024.94	0.6255293	0.2421823	1.615671	-0.4691570
B_2024.94/Bmsy	1.0769481	0.5293546	2.191002	0.0741312
F_2024.94/Fmsy	0.6773982	0.2627482	1.746419	-0.3894960

Predictions w 95% CI (inp\$msytype: s)

	prediction	cilow	ciupp	log.est
B_2026.00	1237.4733351	541.1491928	2829.793106	7.1208269
F_2026.00	0.6255296	0.2021446	1.935680	-0.4691566



B_2026.00/Bmsy	1.2599078	0.5781412	2.745640	0.2310385
F_2026.00/Fmsy	0.6773985	0.2192458	2.092942	-0.3894956
Catch_2025.00	726.1672291	432.4907655	1219.260356	6.5877803
E(B_inf)	1356.4073603	NA	NA	7.2125948



Table.D.4. Summary of main results for case 4.

Convergence: 0 MSG: relative convergence (4)
 Objective function at optimum: 48.1992091
 Euler time step (years): 1/16 or 0.0625
 Nobs C: 24, Nobs I1: 9, Nobs I2: 13, Nobs I3: 9, Nobs I4: 15,
 Nobs I5: 17, Nobs I6: 24

Priors

logr ~ dnorm[log(1), 0.5^2]
 logn ~ dnorm[log(2), 2^2]
 logalpha ~ dnorm[log(1), 2^2]
 logbeta ~ dnorm[log(1), 2^2]

Model parameter estimates w 95% CI

	estimate	cilow	ciupp	log.est
alpha1	0.3639968	0.0841896	1.5737533	-1.0106103
alpha2	0.2933717	0.0621559	1.3846937	-1.2263149
alpha3	1.3638109	0.6943822	2.6786113	0.3102829
alpha4	1.0356958	0.5622894	1.9076754	0.0350734
alpha5	1.8388758	1.0291447	3.2857032	0.6091544
alpha6	1.4775050	0.8532586	2.5584520	0.3903549
beta	0.2465781	0.0542974	1.1197731	-1.4000766
r	1.7121325	0.9282417	3.1580111	0.5377397
rc	2.9508251	1.5569144	5.5927087	1.0820848
rold	10.6712677	0.1153142	987.5278881	2.3675549
m	912.3876184	815.6669113	1020.5773393	6.8160649
K	1563.3258155	939.2819022	2601.9745505	7.3545708
q1	0.0046685	0.0026876	0.0081094	-5.3669163
q2	0.0087009	0.0048971	0.0154594	-4.7443249
q3	0.0163196	0.0087522	0.0304299	-4.1153884
q4	0.0094053	0.0052709	0.0167826	-4.6664833
q5	0.0021436	0.0011701	0.0039271	-6.1452555
q6	0.0008719	0.0004929	0.0015424	-7.0448155
n	1.1604432	0.5973489	2.2543414	0.1488020
sdb	0.2539815	0.1587031	0.4064609	-1.3704937
sdf	0.3345663	0.2131690	0.5250980	-1.0949202
sdi1	0.0924485	0.0260963	0.3275067	-2.3811040
sdi2	0.0745110	0.0180305	0.3079164	-2.5968086
sdi3	0.3463828	0.2147955	0.5585827	-1.0602108
sdi4	0.2630476	0.1767637	0.3914494	-1.3354202
sdi5	0.4670405	0.3285136	0.6639812	-0.7613393
sdi6	0.3752590	0.2745958	0.5128241	-0.9801388
sdC	0.0824967	0.0220875	0.3081245	-2.4949968

Deterministic reference points (Drp)

	estimate	cilow	ciupp	log.est
Bmsyd	618.394919	324.8295998	1177.270408	6.4271273
Fmsyd	1.475413	0.7784572	2.796354	0.3889376
MSYd	912.387618	815.6669113	1020.577339	6.8160649

Stochastic reference points (Srp)



	estimate	cilow	ciupp	log.est	rel.diff.Drp
Bmsys	580.996691	279.0160692	1209.812598	6.3647451	-0.06436909
Fmsys	1.493292	0.7623406	2.925099	0.4009832	0.01197338
MSys	868.266528	774.3296244	973.599278	6.7664987	-0.05081515

States w 95% CI (inp\$msytype: s)

	estimate	cilow	ciupp	log.est
B_2024.94	168.2786474	34.9080365	811.208707	5.1256212
F_2024.94	3.0890019	0.9734955	9.801723	1.1278480
B_2024.94/Bmsy	0.2896379	0.0663318	1.264704	-1.2391238
F_2024.94/Fmsy	2.0685850	0.8050212	5.315442	0.7268648

Predictions w 95% CI (inp\$msytype: s)

	prediction	cilow	ciupp	log.est
B_2026.00	177.4938187	15.2131563	2070.842832	5.178936
F_2026.00	3.0890025	0.8104679	11.773368	1.127848
B_2026.00/Bmsy	0.3054988	0.0264109	3.533748	-1.185809
F_2026.00/Fmsy	2.0685854	0.6479329	6.604149	0.726865
Catch_2025.00	537.1625471	179.5013141	1607.473480	6.286301
E(B_inf)	185.7105654	NA	NA	5.224189



Table.D.5. Summary of main results for case 5.

Convergence: 0 MSG: both X-convergence and relative convergence (5)

Objective function at optimum: 12.7392955

Euler time step (years): 1/16 or 0.0625

Nobs C: 24, Nobs I1: 24

Priors

```
logr ~ dnorm[log(1), 0.5^2]
logn ~ dnorm[log(2), 2^2]
logalpha ~ dnorm[log(1), 2^2]
logbeta ~ dnorm[log(1), 2^2]
```

Fixed parameters

```
fixed.value
n          2
```

Model parameter estimates w 95% CI

	estimate	cilow	ciupp	log.est
alpha	0.3352631	0.0365360	3.0764513	-1.0928398
beta	0.4858885	0.1341091	1.7604153	-0.7217761
r	1.1702071	0.6313067	2.1691275	0.1571808
rc	1.1702071	0.6313067	2.1691275	0.1571808
rold	1.1702071	0.6313067	2.1691275	0.1571808
m	949.7862765	768.3409809	1174.0802501	6.8562370
K	3246.5578191	1732.5095618	6083.7399720	8.0853506
q	0.0004110	0.0002125	0.0007949	-7.7968590
sdb	0.2010545	0.1225708	0.3297923	-1.6041793
sdf	0.2855782	0.1545869	0.5275670	-1.2532393
sdi	0.0674061	0.0101818	0.4462456	-2.6970191
sdc	0.1387592	0.0596742	0.3226538	-1.9750154

Deterministic reference points (Drp)

	estimate	cilow	ciupp	log.est
Bmsyd	1623.2789096	866.2547809	3041.869986	7.3922034
Fmsyd	0.5851036	0.3156534	1.084564	-0.5359664
MSYd	949.7862765	768.3409809	1174.080250	6.8562370

Stochastic reference points (Srp)

	estimate	cilow	ciupp	log.est	rel.diff.Drp
Bmsys	1567.259481	832.1322579	2951.817163	7.357084	-0.03574356
Fmsys	0.576726	0.3081763	1.079294	-0.550388	-0.01452607
MSYs	903.409979	737.3128490	1106.924411	6.806177	-0.05133472

States w 95% CI (inp\$msytype: s)

	estimate	cilow	ciupp	log.est
B_2024.94	1308.2385113	588.7094293	2907.186325	7.1764369
F_2024.94	0.4947634	0.1907482	1.283319	-0.7036756
B_2024.94/Bmsy	0.8347300	0.5053881	1.378691	-0.1806470
F_2024.94/Fmsy	0.8578830	0.4009397	1.835596	-0.1532876

Predictions w 95% CI (inp\$msytype: s)



	prediction	cilow	ciupp	log.est
B_2026.00	1525.0641387	629.6614265	3693.763869	7.3297917
F_2026.00	0.4947637	0.1623794	1.507525	-0.7036751
B_2026.00/Bmsy	0.9730770	0.5006418	1.891330	-0.0272921
F_2026.00/Fmsy	0.8578834	0.3302202	2.228707	-0.1532871
Catch_2025.00	704.6573448	413.0929451	1202.010297	6.5577116
E(B_inf)	1740.1976477	NA	NA	7.4617540



Table.D.6. Summary of main results for case 6.

Convergence: 0 MSG: relative convergence (4)
 Objective function at optimum: 48.2814055
 Euler time step (years): 1/16 or 0.0625
 Nobs C: 24, Nobs I1: 9, Nobs I2: 13, Nobs I3: 9, Nobs I4: 15,
 Nobs I5: 17, Nobs I6: 24

Priors

logr ~ dnorm[log(1.5), 0.15^2]
 logn ~ dnorm[log(2), 2^2]
 logalpha ~ dnorm[log(1), 2^2]
 logbeta ~ dnorm[log(1), 2^2]

Fixed parameters

fixed.value
 n 2

Model parameter estimates w 95% CI

	estimate	cilow	ciupp	log.est
alpha1	0.4814647	0.1555598	1.4901552	-0.7309223
alpha2	0.3434017	0.0842587	1.3995560	-1.0688543
alpha3	1.5241611	0.8110932	2.8641185	0.4214441
alpha4	1.1609344	0.6601668	2.0415579	0.1492252
alpha5	2.0144996	1.1935123	3.4002236	0.7003708
alpha6	1.6418758	1.0061816	2.6791944	0.4958394
beta	0.2857620	0.0723306	1.1289814	-1.2525960
r	1.6768003	1.2722470	2.2099950	0.5168874
rc	1.6768003	1.2722470	2.2099950	0.5168874
rold	1.6768003	1.2722470	2.2099950	0.5168874
m	931.2739643	811.7874403	1068.3476407	6.8365535
K	2221.5500299	1670.5989818	2954.2006125	7.7059604
q1	0.0029543	0.0021447	0.0040694	-5.8244989
q2	0.0052811	0.0039223	0.0071106	-5.2436194
q3	0.0098770	0.0067592	0.0144328	-4.6175482
q4	0.0057707	0.0041750	0.0079762	-5.1549623
q5	0.0013132	0.0009125	0.0018899	-6.6353066
q6	0.0005405	0.0003888	0.0007514	-7.5229971
sdb	0.2284299	0.1535962	0.3397234	-1.4765260
sdf	0.3077498	0.1960702	0.4830409	-1.1784682
sdi1	0.1099809	0.0407640	0.2967273	-2.2074483
sdi2	0.0784432	0.0220333	0.2792739	-2.5453803
sdi3	0.3481639	0.2161620	0.5607745	-1.0550819
sdi4	0.2651921	0.1785708	0.3938317	-1.3273008
sdi5	0.4601719	0.3243252	0.6529193	-0.7761552
sdi6	0.3750535	0.2751685	0.5111964	-0.9806866
sdC	0.0879432	0.0280974	0.2752568	-2.4310642

Deterministic reference points (Drp)

	estimate	cilow	ciupp	log.est
Bmsyd	1110.7750149	835.2994909	1477.100306	7.0128133



```
Fmsyd      0.8384002    0.6361235      1.104998 -0.1762598
MSYd      931.2739643 811.7874403 1068.347641  6.8365535
Stochastic reference points (Srp)
      estimate      cilow      ciupp      log.est rel.diff.Drp
Bmsys 1059.5399275 790.2919621 1420.519139  6.9655901 -0.048355976
Fmsys   0.8321508    0.6255354      1.107012 -0.1837416 -0.007509904
MSYs   881.3768558 768.4009712 1010.963275  6.7814853 -0.056612683
```

```
States w 95% CI (inp$msytype: s)
      estimate      cilow      ciupp      log.est
B_2024.94      244.4767119 79.1390549 755.2385200  5.4991201
F_2024.94       1.8581841  0.8579870  4.0243596  0.6195997
B_2024.94/Bmsy   0.2307386  0.0792664  0.6716626 -1.4664700
F_2024.94/Fmsy   2.2329895  1.0834798  4.6020629  0.8033413
```

```
Predictions w 95% CI (inp$msytype: s)
      prediction      cilow      ciupp      log.est
B_2026.00      166.4831746 15.9220729 1740.768779  5.1148943
F_2026.00       1.8581845  0.6891943  5.009980  0.6195999
B_2026.00/Bmsy   0.1571278  0.0157727  1.565305 -1.8506958
F_2026.00/Fmsy   2.2329900  0.8604063  5.795220  0.8033415
Catch_2025.00  374.8223542 141.2227341  994.824227  5.9264522
E(B_inf)         0.0000000      NA      NA      NA
```



Table.D.7. Summary of main results for case 7.

Convergence: 0 MSG: relative convergence (4)
 Objective function at optimum: 46.6676603
 Euler time step (years): 1/16 or 0.0625
 Nobs C: 24, Nobs I1: 9, Nobs I2: 13, Nobs I3: 9, Nobs I4: 15,
 Nobs I5: 17, Nobs I6: 24

Priors

logr ~ dnorm[log(1.5), 0.15^2]
 logn ~ dnorm[log(2), 2^2]
 logalpha ~ dnorm[log(1), 2^2]
 logbeta ~ dnorm[log(1), 2^2]

Model parameter estimates w 95% CI

	estimate	cilow	ciupp	log.est
alpha1	0.3727460	0.0883848	1.5719856	-0.9868581
alpha2	0.3079943	0.0700658	1.3538777	-1.1776739
alpha3	1.3797918	0.7054488	2.6987434	0.3219326
alpha4	1.0455104	0.5685314	1.9226590	0.0445052
alpha5	1.8700068	1.0614888	3.2943590	0.6259421
alpha6	1.4910415	0.8640396	2.5730357	0.3994749
beta	0.2512924	0.0580372	1.0880586	-1.3811381
r	1.5894054	1.2028116	2.1002536	0.4633600
rc	2.8490687	1.5804429	5.1360236	1.0469922
rold	13.7329303	0.0486698	3874.9579610	2.6197966
m	906.2394268	818.8122379	1003.0014950	6.8093035
K	1638.7685082	1124.9705988	2387.2288096	7.4017003
q1	0.0045207	0.0027395	0.0074598	-5.3990985
q2	0.0084623	0.0049285	0.0145301	-4.7721326
q3	0.0158513	0.0088057	0.0285341	-4.1445062
q4	0.0091310	0.0053253	0.0156565	-4.6960775
q5	0.0020830	0.0011775	0.0036848	-6.1739566
q6	0.0008470	0.0004977	0.0014415	-7.0738020
n	1.1157368	0.6174400	2.0161774	0.1095150
sdb	0.2506347	0.1584703	0.3964006	-1.3837589
sdf	0.3331954	0.2133634	0.5203291	-1.0990261
sdi1	0.0934231	0.0267358	0.3264485	-2.3706169
sdi2	0.0771941	0.0196483	0.3032790	-2.5614327
sdi3	0.3458237	0.2141639	0.5584229	-1.0618263
sdi4	0.2620412	0.1758600	0.3904559	-1.3392537
sdi5	0.4686885	0.3293932	0.6668899	-0.7578168
sdi6	0.3737067	0.2733035	0.5109950	-0.9842840
sdC	0.0837295	0.0233085	0.3007751	-2.4801642

Deterministic reference points (Drp)

	estimate	cilow	ciupp	log.est
Bmsyd	636.165382	345.3599143	1171.839510	6.455459
Fmsyd	1.424534	0.7902215	2.568012	0.353845
MSYd	906.239427	818.8122379	1003.001495	6.809304



Stochastic reference points (Srp)

	estimate	cilow	ciupp	log.est	rel.diff.Drp
Bmsys	604.807993	320.5224142	1141.239091	6.4049110	-0.051846849
Fmsys	1.433855	0.7976914	2.577361	0.3603663	0.006500149
MSys	867.498974	779.0625857	965.974343	6.7656143	-0.044657636

States w 95% CI (inp\$msytype: s)

	estimate	cilow	ciupp	log.est
B_2024.94	181.898450	44.8342724	737.985574	5.2034486
F_2024.94	2.926637	1.0057005	8.516656	1.0738541
B_2024.94/Bmsy	0.300754	0.0742599	1.218060	-1.2014625
F_2024.94/Fmsy	2.041098	0.7972328	5.225675	0.7134877

Predictions w 95% CI (inp\$msytype: s)

	prediction	cilow	ciupp	log.est
B_2026.00	196.3452425	22.3783603	1722.711300	5.2798746
F_2026.00	2.9266378	0.8280108	10.344321	1.0738543
B_2026.00/Bmsy	0.3246406	0.0344015	3.063575	-1.1250365
F_2026.00/Fmsy	2.0410981	0.6422608	6.486589	0.7134879
Catch_2025.00	558.1785248	214.5660276	1452.062422	6.3246788
E(B_inf)	203.9577798	NA	NA	5.3179130